Paleomagnetism of Cretaceous volcanic rocks of the Sverdrup Basin—magnetostratigraphy, paleolatitudes, and rotations[1]

P. J. WYNNE AND E. IRVING
Pacific Geoscience Centre, Geological Survey of Canada, P.O. Box 6000, Sidney, B.C., Canada V8L 4B2

AND

K. G. OSADETZ
Institute of Sedimentary and Petroleum Geology, Geological Survey of Canada, 3303-33rd St. NW, Calgary, Alta., Canada T2L 2A7

Received September 23, 1987

Revision accepted February 24, 1988

The principal magnetization of lavas of the Isachsen and Strand Fiord formations on Axel Heiberg Island is shown to predate the Eocene Eurekan Orogeny. Basalt flows of the Strand Fiord Formation, volcanigenic sandstone from the Christopher Formation, and the uppermost flows of the Isachsen Formation are normally magnetized. Reversed magnetizations are found only in the Isachsen Formation, occurring at two horizons, which, we suggest, correspond to M_0 and M_1 of the M sequence of marine magnetic anomalies (118−123 Ma). It is possible, therefore, that we have located, at least approximately, the base of the Cretaceous normal polarity superchron in these sections. Because inclinations are steep, the analysis of directions of magnetization is not straightforward and has been done by two methods. Method I assumes that no relative rotations have occurred amongst sample localities, and calculations on this basis show a 33 ± 24° ($P = 0.05$) counterclockwise rotation with no paleolatitudinal displacement relative to North America. The rotation is in agreement with the rotation of 36 ± 8° ($P = 0.05$) determined earlier from the Permian Esayoo Formation on Ellesmere Island. Analysis by method I assumes that the Esayoo and the Isachsen − Strand Fiord sampling localities on Axel Heiberg and Ellesmere Island are contained within what is essentially one large structural domain. The agreement (using method I) of paleolatitude with that of North America is consistent with standard plate reconstructions in which there is a gap of about 300 km between Greenland and Ellesmere Island. However, the dispersion of site-mean directions is greater than that expected for paleosecular variation during the Cretaceous, and therefore some of the dispersion may be attributable to relative motions amongst collecting localities. Therefore, by method II, relative rotations amongst localities are assumed to have occurred, and inclinations and declinations are analysed separately. As with method I, declinations are predominantly counterclockwise from that expected, but by method II the mean inclination (74 ± 2° standard error) is significantly shallower than that expected (79 ± 1° standard error). This apparent flattening is consistent with the idea that the Arctic Islands were close to Greenland in the Cretaceous and that there was no gap along Nares Strait. Hence both methods of calculation yield similar counter clockwise rotation, but each gives slightly different paleolatitudes. The latter difference cannot at present be resolved.

Il est démontré que l'acquisition de l'aimantation prédominante dans les laves des formations d'Isachsen et de Strand Fiord sur l'île Axel Heiberg date d'avant l'orogenèse de l'Eureka de l'Éocène. Les coulées de laves de la Formation de Strand Fiord, le grès volcanogénique de la Formation de Christopher et les coulées du sommet de la Formation d'Isachsen possèdent une aimantation normale. Les seules aimantations inversées apparaissent dans la Formation d'Isachsen, elles correspondent à deux horizons désignés M_0 et M_1 dans la séquence M des anomalies magnétiques marines (118−123 Ma). Il est plausible que nous ayons localisé dans ces coupes, au moins de manière approximative, la base du superchrone normale crétacée. Vu que les inclinaisons sont abruptes, il a fallu avoir recours à deux méthodes d'analyse des directions d'aimantation. La première méthode écarte toute possibilité de rotations relatives aux localités échantillonnées, et les calculs fondés sur cette hypothèse indiquent une rotation de sens anti-horaire de 33 ± 24° ($P = 0{,}05$) et sans déplacement de la paléolatitude relativement à l'Amérique du Nord. Cette rotation s'accord avec celle de 36 ± 8° ($P = 0{,}05$) déterminée aùparavant pour la Formation d'Esayoo sur l'île Ellesmere. Dans l'application de la méthode I, on considère que les sites d'échantillonnage des formations d'Essayoo et d'Isachsen − Strand Fiord sur les îles Axel Heiberg et Ellesmere font partie d'un même grand domaine structural. La concordance (avec la méthode I) de paléolatitude avec celle de l'Amérique du Nord est conforme aux reconstructions conventionnelles des plaques pour lesquelles existe un hiatus d'environ 300 km entre le Groenland et l'île Ellesmere. Cependant, la dispersion des directions moyennes par sites est plus grande que celle estimée pour la variation paléoséculaire durant le Crétacé, et par conséquent une part de cette dispersion peut être attribuée aux mouvements relatifs parmi les localités rassemblées. Dans la méthode II, on suppose qu'il y a eu des rotations relatives parmi les localités, et dans ce cas les inclinaisons et les déclinaisons ont été analysées séparément. Comme pour la méthode I, les déclinaisons sont principalement dans le sens anti-horaire comme attendue, mais la méthode II révèle que l'inclinaison moyenne (74 ± 2° d'écart-type) est significativement moins prononcée que l'angle attendu (79 ± 1° d'écart-type). Cette tendance vers l'horizontal est compatible avec l'hypothèse que les îles de l'Arctique étaient proches du Groenland durant le Crétacé et qu'il n'existait pas d'hiatus le long du détroit de Nares. Ainsi, les deux méthodes de calcul indiquent une rotation dans le sens anti-horaire mais avec chacune une paléolatitude légèrement différente. On ne peut expliquer cette dernière différence.

[Traduit par la revue]

Can. J. Earth Sci. **25**, 1220−1239 (1988)

[1]Geological Survey of Canada Contribution 42487.

Printed in Canada / Imprimé au Canada

Introduction

All previous paleomagnetic investigations of Cretaceous rocks of the Sverdrup Basin have been carried out on intrusive igneous bodies. Work by Larochelle and Black (1963) on dykes and sills intrusive into flat-lying Early Cretaceous sedimentary rocks near the Isachsen weather station on Ellef Ringnes Island (Fig. 1) yielded a paleopole (69°N, 180°W, A_{95} = 14°) that is commonly used as one of the cratonic reference poles. Wynne *et al.* (1983) reported magnetizations from a sill and its adjacent wall rock on Ellesmere Island, which they interpreted as Cretaceous (D, I, α_{95}, 352, 81, 6° and 035, 83, 10° for the sill and the wall rock, respectively, when corrected for the tilt of enclosing beds). Jackson and Halls (1985) studied sills on eastern Axel Heiberg and identified a primary, pre-orgenic, syndiapiric, early Cretaceous magnetization.

The present work is concerned mainly with extrusive mafic lavas. They are intercalated with sedimentary rocks, and their attitude, general structure, and age can be determined. Originally the work had two objectives. The first was to make comparisons with Cretaceous data from cratonic North America in order to determine if time-correlative rocks of Axel Heiberg and Ellesmere islands (Fig. 1) had been rotated or moved latitudinally relative to the craton during the Eurekan Orogeny. The second objective was to tie the sequence into the youngest of the M sequence of geomagnetic reversals.

Because many of our arguments are based on comparisons with the cratonic paleofield, the cratonic data are reviewed at the outset. Currently the middle Cretaceous (100 Ma) reference field for cratonic North America is based on studies from five areas yielding well-grouped paleopoles (Fig. 2). The rock units range in age from about 78 to 130 Ma (see caption to Fig. 2) during which the configuration of the geomagnetic field remained essentially constant and of predominantly normal polarity. The paleopoles have been averaged to obtain the cratonic reference paleopole for this interval of time (see Table 4). The Ellef Ringnes Island data (C of Fig. 2, situated about 150 km west of our study area) have been omitted because the error is larger than is acceptable by current standards for cratonic data. It is notable that paleomagnetic studies (reviewed by Vincenz and Jelenska 1985) of Mesozoic dolerites of Svalbard, which are regarded as coeval with the Cretaceous volcanics of the Queen Elizabeth Islands (Vincenz and Jelenska 1985), have yielded a paleopole concordant with those from cratonic North America after correction for closure of the North Atlantic (SV of Figs. 2 and 15).

An anticlockwise rotation of 36 ± 8° within the Sverdrup Basin (Fig. 1) has been observed previously in Permian volcanics of the Esayoo Formation of Ellesmere Island (Wynne *et al.* 1983). The inclination of the Permian magnetization is −62° as opposed to the very steep inclinations (~80°) for Cretaceous magnetizations. Accordingly, the vertical-axis rotations in the Sverdrup Basin can be much more accurately determined from Permian units. The Esayoo rotation was thought to have been produced during the Eurekan Orogeny, considered, in the main, as being Middle to Late Eocene in age (Ricketts and McIntyre 1986; Embry and Osadetz 1988), contemporaneous, in part, with the later stages of the opening of the Labrador Sea and Baffin Bay. If the rotation of the Esayoo volanics was indeed Eurekan, then similar rotations might be expected to have affected Cretaceous rocks of the Strand Fiord and Isachsen formations exposed to the south and west on Axel Heiberg Island.

In the standard reconstruction of Greenland and North America there is a sinistral offset of 250 km and a gap of about 300 km between Ellesmere Island and Greenland, yet most workers would argue that a gap or offset of such magnitude did not exist because of geological continuity across Nares Strait (Dawes and Kerr 1982). Attempts to reconcile this conflict have appealed either to deformation distributed across Queen Elizabeth Islands (Keen and Peirce 1982) or to motion of the Queen Elizabeth Islands as a whole relative to Greenland and North America, with some motion occurring along Nares Strait and some along the Northwest Passage (Hamilton 1983). As can be seen in Fig. 2, the Cretaceous reference paleopole is situated west of Ellesmere and Axel Heiberg islands, so the differences between the two hypotheses (gap or no gap) would be expressed as differences in paleolatitude. In principle, therefore, paleomagnetic studies could decide this issue because from them paleolatitude can be estimated. Moreover, if there had been substantial strain across the Queen Elizabeth Islands, the shortening required would have been taken up within the Eurekan Orogen, and paleomagnetic data collected across the orogen could, in principle, document this shortening. In practice, however, the latitude differences expected are small (3−4°) and at or beyond the limits of resolution of the paleomagnetic method. Furthermore, the problem is exacerbated by the structural complexities of the region and by the fact that because the paleolatitude is high the inclinations are steep and paleolatitude is difficult to determine.

In the area sampled, volcanic rocks of the Isachsen Formation range in age from Barremian to Aptian. During this time interval the youngest of the M sequence of reversals of the geomagnetic field occurred. These reversals have been documented in carbonate sections in Italy (Lowrie *et al.* 1980), but carbonates are not amenable to isotopic dating, whereas the Isachsen basalts are. Our second objective, therefore, was to identify horizons of reversed magnetization corresponding to these anomalies in rocks that could be both paleontologically and isotopically dated. This would refine the reversal time scale and expedite its correlation to the biostratigraphic time scale for the Early Cretaceous, a key interval of time.

Geology and sampling

The Sverdrup Basin (Fig. 1) contains a 13 km thick sequence of Lower Carboniferous to lower Tertiary carbonate, evaporite, clastic, volcanic, and intrusive rocks (Balkwill 1978). Four episodes of Cretaceous volcanism occurred during a sequence of transgressive−regressive cycles of clastic sedimentation (Embry and Ozadetz 1988). These episodes are Valanginian − early Barremian, late Barremian − Aptian, latest Aptian − early Cenomanian, and Late Cenomanian − Maastrichtian in age. During the first three episodes, quartz tholeiites were produced and formed part of a volcanic province that today extends to the Svalbard and Franz Josef archipelagos of the Barents Shelf (Embry and Ozadetz 1988). Within the Queen Elizabeth Islands the lavas increase in cumulative thickness northward, and their volume and areal extent increase in each succeeding episode. During the fourth volcanic episode a greater diversity of volcanic rock types was produced, including basalt, alkali basalt, trachyte, and rhyolite. Volcanic rocks from episodes 2 and 3 have been sampled.

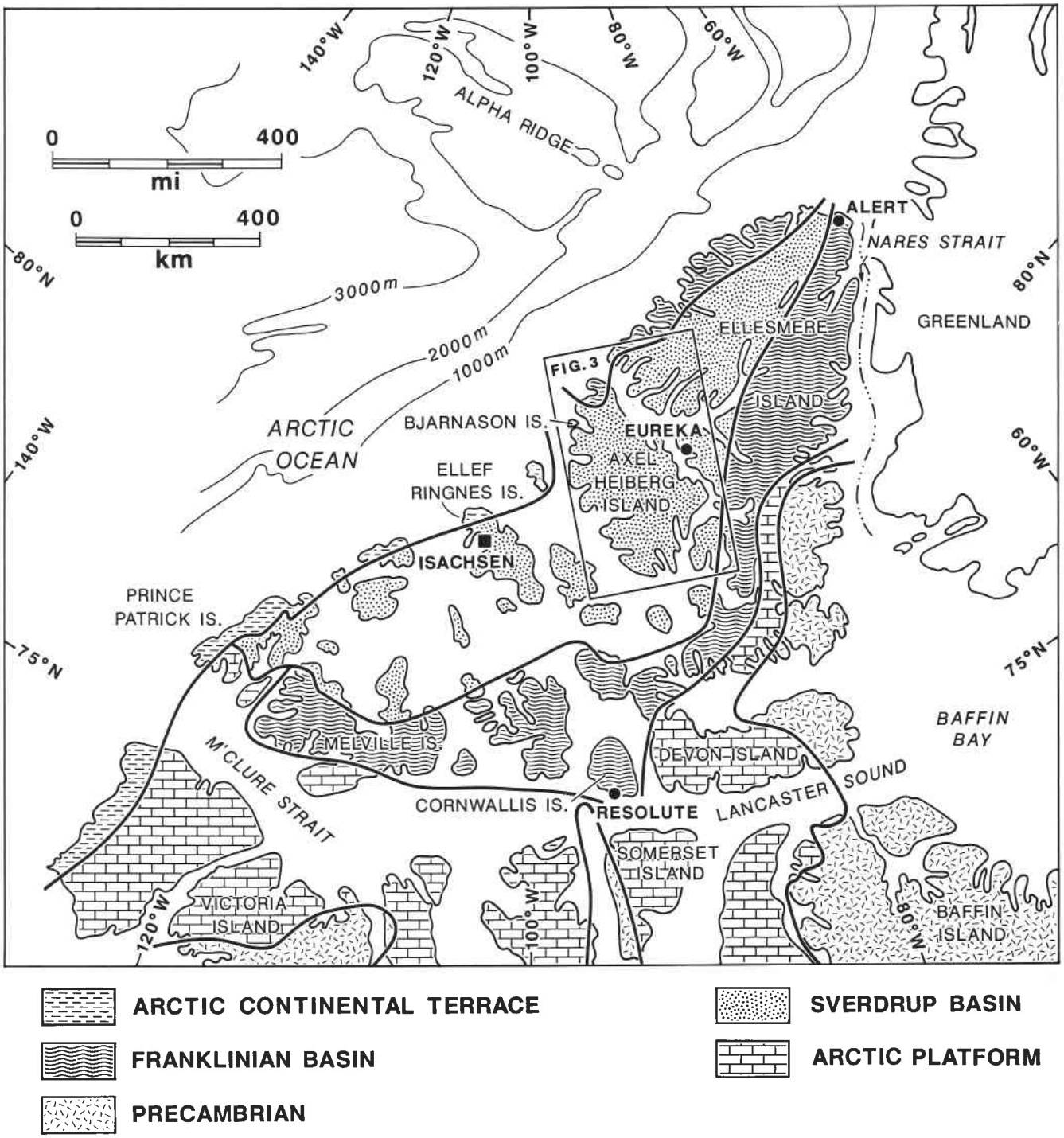


FIG. 1. Sketch map of the Queen Elizabeth Islands showing the Sverdrup Basin. The sampling location of the Isachsen diabase study (Larochelle and Black 1963) is shown. Bathymetry from Weber (1986).

Lavas of episodes 1 and 4 were not sampled for this study, the former because they are thin and rare, the latter because their age and correlation have been recognized only recently.

Volcanic rocks of episode 2, entirely subaerial, are quartz tholeiite lavas, epiclastic breccias, and lesser pyroclastics. They occur interbedded with fluvial quartzose sandstones of the Walker Island Member, and equivalent strata, in the upper portion of the Isachsen Formation (Valanginian–Aptian). Volcanic rocks of episode 3 are contained within the Strand Fiord Formation (latest Aptian – early Cenomanian). They occur as thick, monotonous quartz tholeiitic lavas, both subaerial and submarine, interbedded at their southern margin with lesser pyroclastic and epiclastic strata (Ricketts *et al.* 1985).

Middle to Late Eocene Eurekan thrusting resulted in the deformation of the Sverdrup Basin (Ricketts and McIntyre 1986). Most Cretaceous volcanic rocks in northwestern Axel Heiberg are preserved in structural depressions, either in the footwall of major thrusts or in thrust-plate synclinoria. In west-central Axel Heiberg the volcanics are preserved in dome

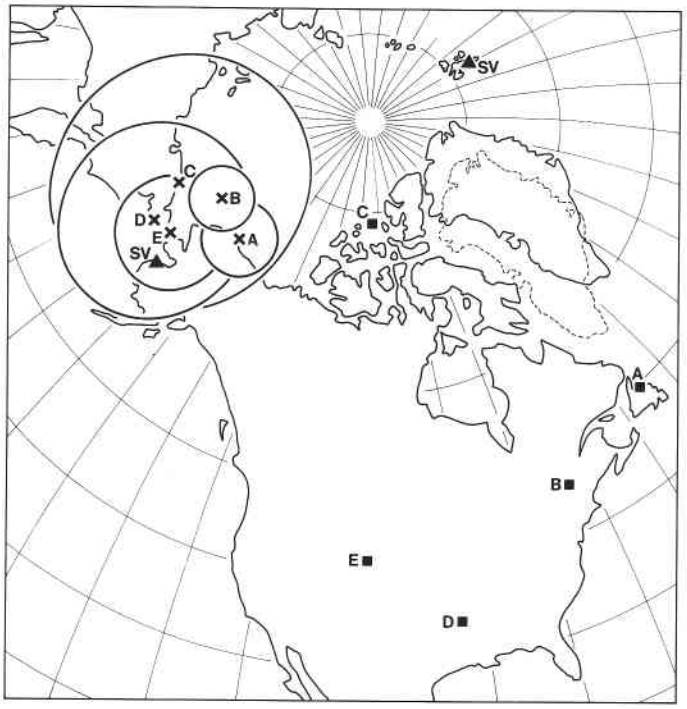


FIG. 2. Cretaceous cratonic paleopoles from North America: A, lamprophyre dykes of Notre Dame Bay, 129 Ma, 71°N, 154°W, A_{95} = 4° (Lapointe 1979; Prasad 1981); B, Monteregian Hills, 120−100 Ma, 72°N, 169°W, A_{95} = 3° (Foster and Symons 1979); C, Isachsen diabase from Ellef Ringnes Island, 101 Ma, 65°N, 180°, A_{95} = 14° (Larochelle and Black 1963); D, alkalic intrusions, Arkansas, 105−95 Ma, 65°N, 173°W, A_{95} = 11° (Scharon and Hsu 1969); E, Niobrara Formation, Wyoming−Colorado−Kansas, Coniacian−Maastrichtian, 66°N, 168°W, A_{95} = 6° (Shive and Frerichs 1974; Gordon *et al.* 1984); SV, Svalbard Mesozoic dolerite paleopole 65°N, 138°W, A_{95} = 12°, rotated as in Fig. 18.

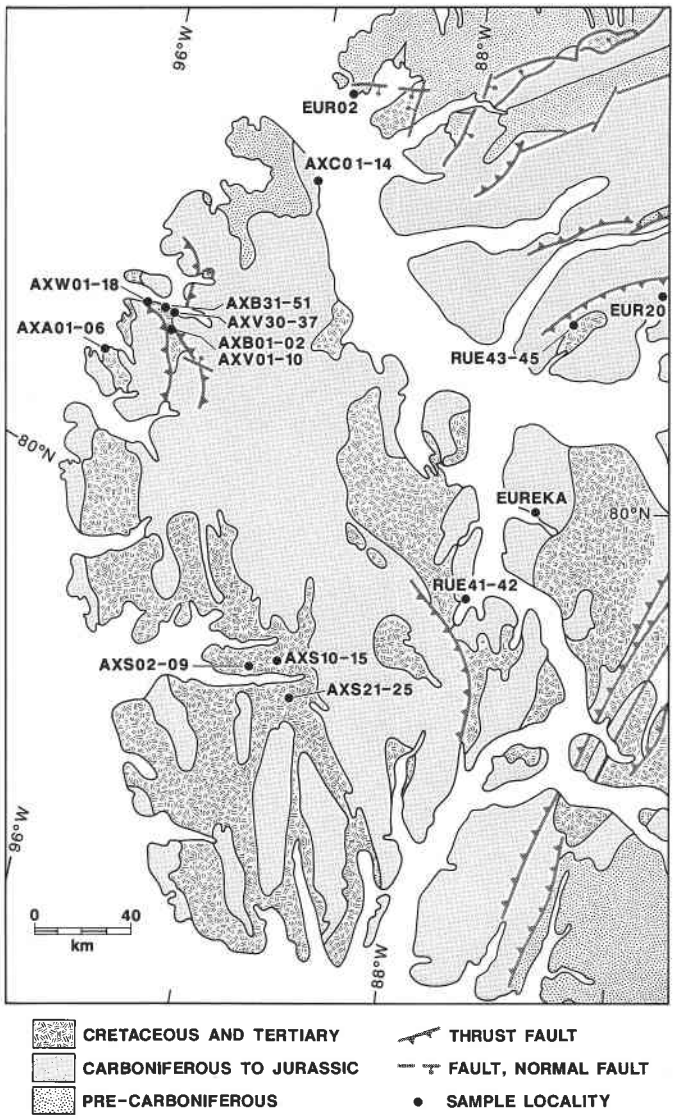


FIG. 3. Geology of sampling regions. Sample localities are listed in Table 1. The two thrust faults shown in northwestern Axel Heiberg are Griesbach Creek (eastern) and Bunde Fiord (western).

and basin structures that have structural culminations commonly cored by diapirs of Carboniferous evaporites. All localities sampled are from detached thrust sheets (Ricketts 1986, 1987; Fischer 1984).

Lava flows and volcaniclastic rocks of the Isachsen and Strand Fiord formations were sampled on Axel Heiberg Island and Ellesmere Island at 14 localities (Fig. 3; Table 1). At each locality, samples were obtained from several sites, generally individual lava flows, in stratigraphic sequence. Several dykes and sills, the hypabyssal equivalents of the flows, were sampled as well. A nodular red sandstone, belonging to the Permian Assistance Formation situated adjacent to one of the sills near Cape Stallworthy, was also sampled (AXC01-14, Fig. 3). Attitudes of flows were obtained from interbedded clastic strata. Attitudes of diabases were obtained similarly on the assumption that they were intruded into flat-lying strata, an assumption that may not always be justified.

Some cores (2.5 cm in diameter and usually five or six per site) were obtained by drilling, but the majority of samples were collected as oriented hand samples, usually three per site. Orientation was by sun compass, or by topographic bearings when the sun was not shining, as it frequently was not. Accurate magnetic bearings were impossible to obtain so near the magnetic dip pole. From each oriented hand sample, usually three cores were drilled in the workshop. Cores drilled and oriented in the field and those taken from oriented hand samples were cut into two or more specimens 2.2 cm in length. One polished thin section was obtained from each site for study in reflected light. A Schonstedt fluxgate spinner magnetometer was used for measurements. A pair of specimens (generally cut from the same core) from each site was demagnetized in detail, one specimen thermally and the other by alternating magnetic field (AF). The behaviour of these pilot specimens then was used to indicate the demagnetization required to define the direction of the principal stable magnetization present in the remaining specimens from that site. In total, 972 specimens from 81 collecting sites at 14 localities were studied.

Descriptions of magnetizations

The majority of lava specimens contain a component of magnetization parallel to the present Earth's field (PEF), which is demagnetized either after heating to low temperatures (200°C) or after low AF treatment (10−20 mT). After removal of the PEF magnetization, the magnetization is characteristically one with moderate coercivity (30−50 mT) and with unblocking temperatures predominantly in the range of

TABLE 1. Site locations with bedding attitudes

Locality	Site No.	Rock unit	Site		Bedding attitude	
			Lat. (N)	Long. (W)	BDDA (°)	BD (°)
Keybolte Peninsula	EUR02	Porphyry dyke	81°35′	91°24′	173	40
Van Royen Ridge	EUR20	Diabase dyke	80°52′	83°12′	163	63
Mokka Ridge	RUE41	Isachsen basalt	79°38′	87°10′	072	38
	RUE42	Isachsen basalt	79°38′	87°10′	072	38
Blue Mountain	RUE43	Isachsen basalt	80°45′	85°05′	130	30
	RUE44	Isachsen basalt	80°41′	85°05′	130	30
	RUE45	Isachsen basalt	80°38′	85°05′	011	71
Bunde Fiord (west)	AXW02	Diabase sill	80°35′	95°20′	224	52
	AXW01	Diabase sill	80°35′	95°20′	304	46
	AXW03	Diabase sill	80°35′	95°20′	221	58
	AXW04	Isachsen basalt (lowest site)	80°35′	95°20′	221	58
	AXW08	Isachsen basalt	80°35′	95°20′	225	62
	AXW09	Isachsen basalt	80°35′	95°20′	223	77
	AXW10	Isachsen basalt	80°35′	95°20′	223	77
	AXW11	Isachsen basalt	80°35′	95°20′	223	77
	AXW12	Isachsen basalt	80°35′	95°20′	223	77
	AXW13	Isachsen tuff	80°35′	95°20′	235	81
	AXW14	Isachsen basalt (highest site)	80°35′	95°20′	235	81
	AXW18	Strand Fiord basalt	80°35′	95°35′	264	30
Celluloid Creek (north section)	AXB44	Diabase dyke	80°35′	94°50′	193	24
	AXB45	Isachsen basalt (lowest site)	80°35′	94°50′	094	21
	AXB46	Isachsen basalt	80°35′	94°50′	094	21
	AXB47	Isachsen basalt (highest site)	80°35′	94°50′	094	21
	AXB50	Christopher volcanic sandstone	80°35′	94°50′	145	67
Celluloid Creek (south section)	AXB49	Isachsen basalt (lowest site)	80°35′	94°50′	194	15
	AXB32	Isachsen basalt	80°35′	94°50′	194	15
	AXB33	Isachsen basalt	80°35′	94°50′	194	15
	AXB31	Isachsen basalt	80°35′	94°50′	194	15
	AXB43	Strand Fiord basalt	80°35′	94°50′	194	31
	AXB42	Strand Fiord basalt	80°35′	94°50′	194	31
	AXB41	Strand Fiord basalt	80°35′	94°50′	194	31
	AXB40	Strand Fiord basalt	80°35′	94°50′	194	31
	AXB39	Strand Fiord basalt	80°35′	94°50′	180	28
	AXB38	Strand Fiord basalt	80°35′	94°50′	180	28
	AXB37	Strand Fiord basalt	80°33′	94°50′	180	28
	AXB35	Strand Fiord basalt (highest site)	80°33′	94°50′	180	28
Camp Five Creek ("pit stop")	AXB01	Isachsen basalt	80°30′	94°37′	223	62
	AXB02	Isachsen basalt	80°30′	94°37′	223	62
Camp Five Creek (south section by the creek)	AXV10	Isachsen basalt (lowest site)	80°28′	94°30′	248	25
	AXV01	Isachsen basalt	80°28′	94°32′	290	15
	AXV02	Isachsen basalt	80°28′	94°32′	290	15
	AXV03	Isachsen basalt	80°28′	94°32′	263	18
	AXV04	Isachsen basalt	80°28′	94°32′	263	17
	AXV05	Isachsen basalt	80°28′	94°32′	263	17
	AXV09	Isachsen basalt (highest site)	80°29′	94°32′	268	22
Camp Five Creek (north section at the fiord)	AXV30	Isachsen basalt (lowest site)	80°33′	94°40′	270	35
	AXV31	Isachsen basalt	80°33′	94°40′	265	40
	AXV32	Isachsen basalt	80°33′	94°40′	255	46
	AXV34	Isachsen basalt	80°33′	94°40′	235	55
	AXV35	Isachsen basalt	80°33′	94°40′	235	55

TABLE 1 (*concluded*)

Locality	Site No.	Rock unit	Site		Bedding attitude	
			Lat. (N)	Long. (W)	BDDA (°)	BD (°)
	AXV36	Isachsen basalt	80°33′	94°40′	235	55
	AXV37	Isachsen basalt (highest site)	80°33′	94°40′	235	55
Bals Fiord	AXA05	Isachsen basalt (lowest site)	80°23′	96°05′	035	26
	AXA06	Isachsen basalt	80°23′	96°05′	035	26
	AXA01	Isachsen basalt	80°23′	96°05′	035	26
	AXA02	Strand Fiord basalt	80°23′	96°05′	035	26
	AXA03	Strand Fiord basalt	80°23′	96°05′	035	26
	AXA04	Strand Fiord basalt (highest site)	80°23′	96°05′	035	26
"Anticline" (north limb)	AXS04	Strand Fiord basalt	79°17′	91°54′	022	54
	AXS02	Strand Fiord basalt	79°17′	91°54′	020	45
"Anticline" (south limb)	AXS07	Strand Fiord basalt	79°16′	91°10′	200	93
	AXS09	Strand Fiord basalt	79°16′	91°10′	200	93
Bastion Ridge	AXS15	Strand Fiord basalt (lowest site)	79°18′	90°40′	335	41
	AXS14	Strand Fiord basalt	79°18′	90°40′	335	41
	AXS13	Strand Fiord basalt	79°18′	90°40′	335	41
	AXS12	Strand Fiord basalt	79°18′	90°40′	335	41
	AXS11	Strand Fiord basalt	79°18′	90°40′	335	41
	AXS10	Strand Fiord basalt (highest site)	79°18′	90°40′	335	41
"South of Camp" (south of Strand Fiord)	AXS25	Strand Fiord basalt (lowest site)	79°08′	90°40′	265	78
	AXS24	Strand Fiord basalt	79°08′	90°40′	265	78
	AXS22	Strand Fiord basalt	79°08′	90°40′	265	78
	AXS21	Strand Fiord basalt (highest site)	79°08′	90°40′	265	78
Cape Stallworthy	AXC01	Diabase dyke	81°14′	92°20′	135	23
	AXC02	Diabase sill	81°14′	92°20′	139	20
	AXC12	Diabase sill	81°12′	92°05′	143	16
	AXC13	Diabase sill	81°12′	92°05′	143	16
	AXC14	Diabase sill	81°12′	92°05′	140	12
	AXC09	Assistance sandstone (lowest site)	81°14′	92°20′	139	20
	AXC10	Assistance sandstone	81°14′	92°20′	139	20
	AXC11	Assistance sandstone	81°14′	92°20′	139	20
	AXC04	Assistant sandstone (highest site)	81°14′	92°20′	139	20

NOTES: BDDA (bedding down-dip azimuth) is the direction, reckoned east of geographic north, in which the beds dip. BD is the magnitude of dip.

300−450°C, properties typical of titanomagnetite, although there is much variability.

Studies of textures and intergrowths in reflected light show that the oxidation state of the iron oxides is generally low. The majority of specimens from both the Isachsen and Strand Fiord formations fall in categories I and II of Wilson *et al.* (1968), their lowest oxidation categories. The titanomagnetite is present as euhedra (50−100 μm), as grains showing skeletal quench textures (up to 100 μm), and as very tiny disseminated particles (10 μm or less). Ilmenite, which is nonmagnetic above −200°C, is present as clean, embayed lathes (30 μm by 100 μm) and rarely as thin lamella or subgrains within titanomagnetite crystals. Hematite was found in only two of the 21 sections examined, occurring as tiny (10 μm or less) elongate blebs along the margins of ilmenite lathes. Titanomagnetite and ilmenite each make up 1−2% of the rock, and together they constitute less than 3%. Except in the intrusions, pyrite and chalcopyrite are present in trace amounts only.

Iron minerals of the Strand Fiord Formation are generally finer grained than those of the Isachsen Formation, and it is only in them that delicate skeletal quench textures may be found. This is consistent with the subaerial character of the Isachsen lavas, compared with the partially submarine character of the Strand Fiord lavas. The more stable magnetization of the Strand Fiord Formation is presumably a consequence of its finer grain size.

Three types of magnetic behaviour are present in both formations.

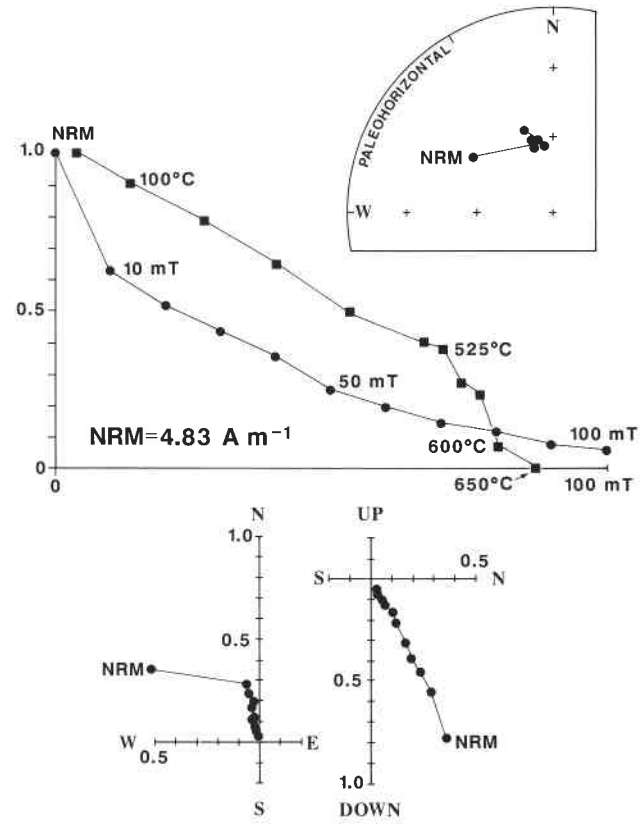


FIG. 4. *A*-type magnetization, Strand Fiord Formation, site ASX10. Changes in direction with AF treatment above right (solid dots denote downward inclination); normalized orthogonal plots below. Central diagram shows intensity changes; squares are thermal, dots are alternating field (AF) demagnetization steps.

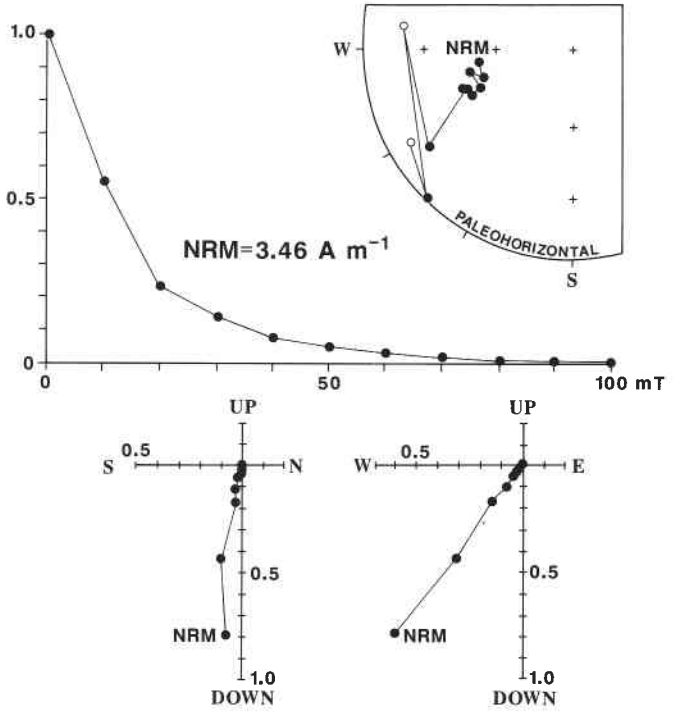


FIG. 5. *B*-type magnetization, Strand Fiord Formation, site AXS21. Typical behaviour of the majority of specimens during AF demagnetization. Changes of direction above right, intensity above left, and normalized orthogonal plots below.

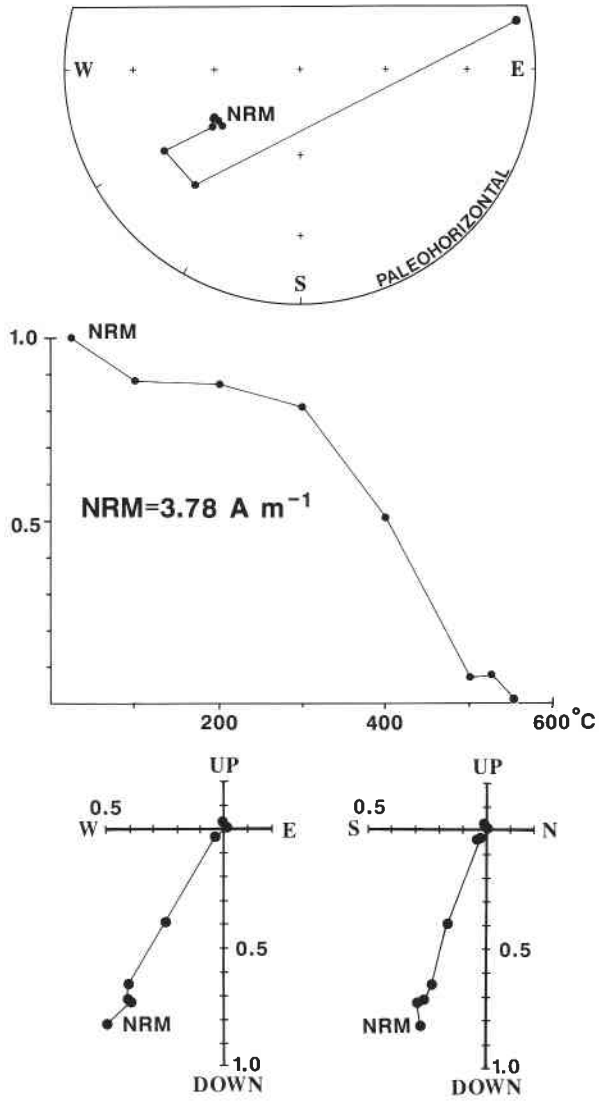


FIG. 6. *B*-type magnetization, Strand Fiord Formation, site AXS21. Typical behaviour of the majority of specimens during thermal demagnetization. Changes of direction above, of intensity in middle, and normalized orthogonal plots below.

A-*type magnetization*

About 10% of the collected samples have essentially "ideal" (*A*-type) magnetizations. After removal of the PEF magnetization, the directions remain constant over the treatment range (up to 550°C and occasionally higher, or 100 mT), forming excellent end points and orthogonal plots that march straight to the origin (Fig. 4). For *A*-type specimens, the direction of stable magnetization has been chosen as that observed at a single demagnetization level (generally between 20 and 60 mT) in the middle of the end-point range. Decay during AF demagnetization is smooth, and at 100 mT about 10% of the magnetization is left (Fig. 4 centre). Although the directions remain constant, the decay of intensity during thermal demagnetization has three steps. In the example shown in Fig. 4 they are (*i*) a gentle slope up to 525°C, (*ii*) a steep fall between 525 and 600°C, and (*iii*) a toe between 600 and 700°C. These steps probably correspond to the three magnetic constituents observed in polished section. The magnetization with distributed blocking temperatures up to 525°C is presumably carried by embayed titanomagnetite euhedra. Very small grains

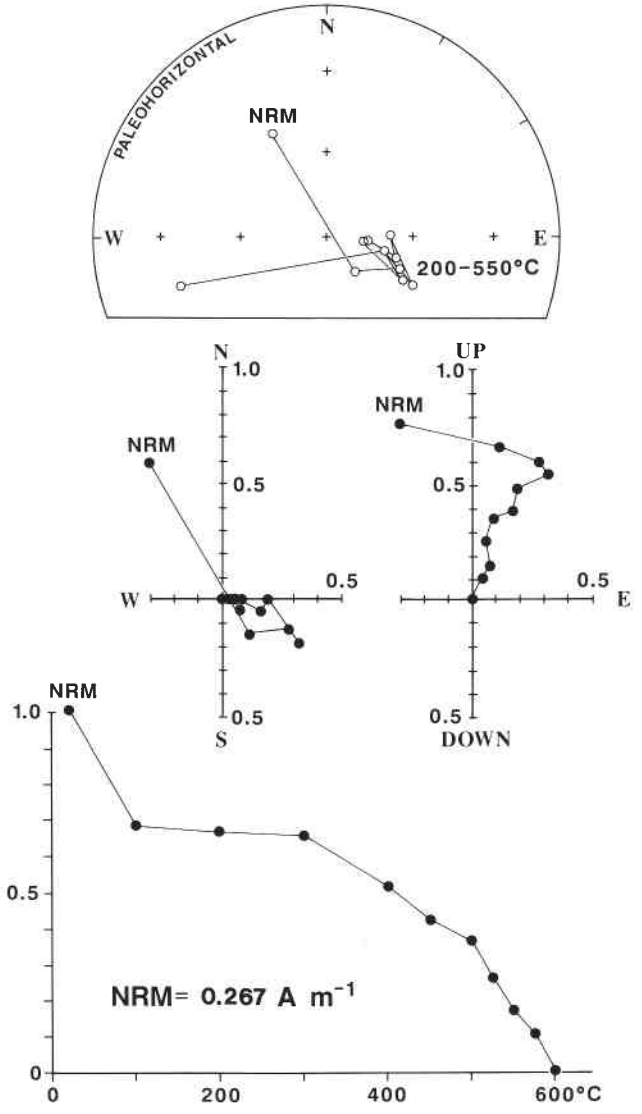


FIG. 7. *B*-type magnetization, Isachsen Formation, site AXV03. Thermal demagnetization of a specimen from a reversely magnetized basalt flow. Changes in direction above (open circles denote upward inclinations), normalized orthogonal plots centre, and changes in intensity below.

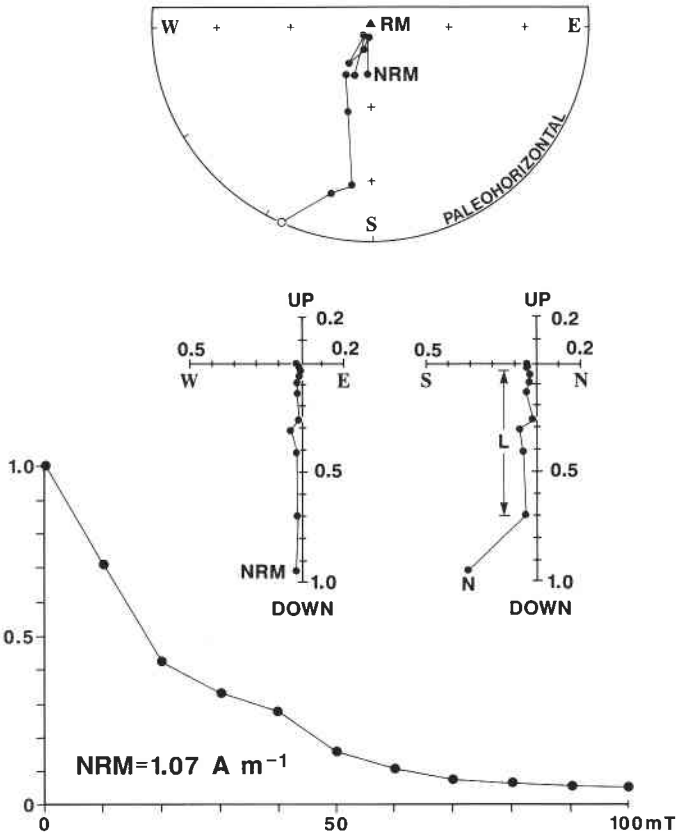


FIG. 8. *B*-type magnetization, Strand Fiord Formation, site AXB42. AF demagnetization showing migration towards paleohorizontal. Changes in direction above (solid dots down, and open circles, up directions), normalized orthogonal plots centre, and changes in intensity below. RM is the direction of the magnetization removed during demagnetization along the approximately straight line segment (L).

of titanomagnetite scattered throughout the polished sections could be responsible for the magnetization with unblocking temperatures between 525 and 600°C. Tiny blebs of hematite, present along the margins of ilmenite lathes, are responsible presumably for the "hematite toe" above 600°C. All three magnetizations have the same direction.

B-*type magnetization*

B-type magnetization characterizes the majority of the collection and is of somewhat lower stability than *A*-type. Trajectories of the orthogonal plots generally are more irregular (Fig. 5, bottom), and end points are less precise. The directions of magnetizations remain roughly consistent up to 60 mT (in some specimens up to 100 mT) and then generally become internally random. Directions of the principal magnetization were calculated either by vector subtraction (between the limits of approximately straight line segments) or by LINEFIND, a computer programme that selects the line that best fits these segments (Kent *et al.* 1983).

Typical examples of the *B*-type magnetization are shown in Figs. 5–7. The specimen depicted in Fig. 5 was obtained from a submarine flow situated near the southern limit of the Strand Fiord volcanic pile where it intertongues with the marine Bastion Ridge Formation. Thermal demagnetization yields end points between 300 and 500°C, and the decay curves often show two steps (Fig. 6). There is a rounded shoulder in the range 300–500°C. Above 500°C only 10% of the original magnetization remains, and there is a small toe ("magnetite toe") that disappears at 550°C. This toe occurs at a lower temperature than the hematite toe of the *A*-type magnetizations and is a common feature of the intensity-decay curves of basalts from both formations. At high temperatures (450–600°C) and high alternating fields (80 mT), where the intensity is less than 10% that of the natural remanent magnetization (NRM), the directions become scattered (Figs. 5 and 6). An example of *B*-type magnetization with reversed polarity is shown in fig. 7.

The basalt of Fig. 5 contains titanomagnetite with delicate skeletal quenched textures, typical of rapidly cooled basalts. The groundmass, interstitial between plagioclase lathes, is peppered with very fine grained titanomagnetite or magnetite less than 10 μm. The principal magnetization, with its moderate coercivity and unblocking temperatures, is probably carried by skeletal titanomagnetite; the magnetite toe, by the very fine grained interstitial titanomagnetite. Large (100 μm)

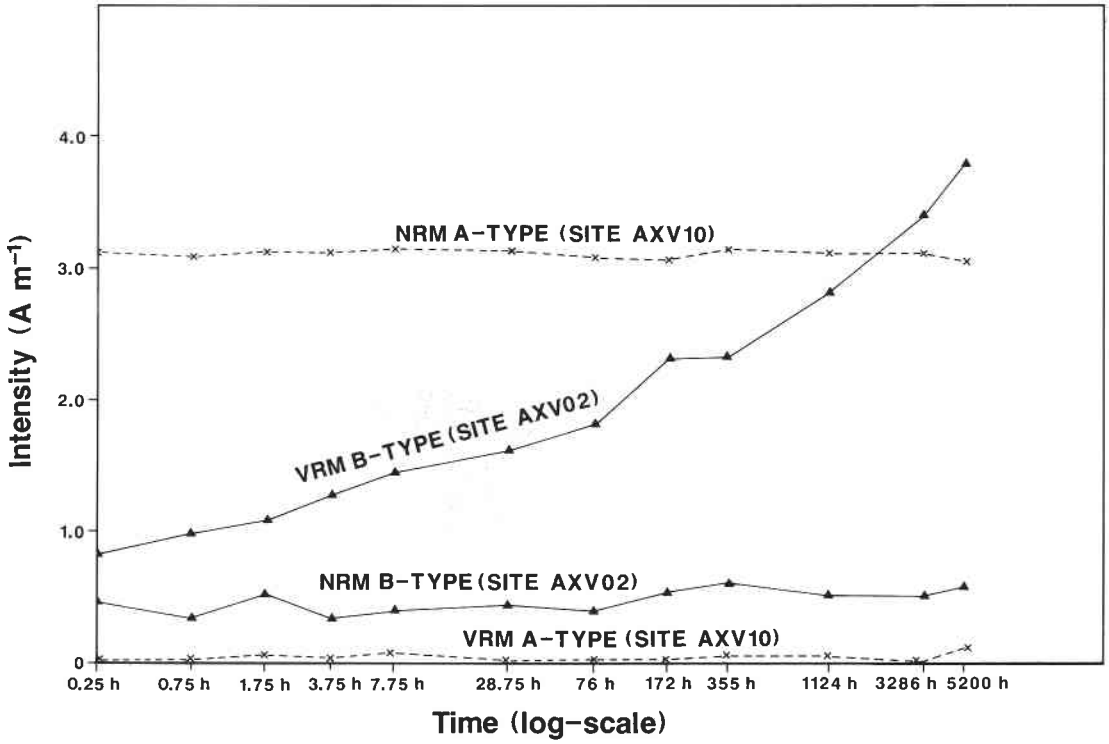


FIG. 9. Viscous decay of remanence and buildup of VRM in *A*- and *B*-type specimens.

euhedra of titanomagnetite in the reversed polarity flows of Camp Five Creek (Fig. 7) have undergone varying degrees of alteration along curved fractures that internally subdivide the crystals.

A handful of *B*-type specimens show a systematic and well-defined migration to paleohorizontal in the temperature range of 450–600°C and in AF above 80 mT (Fig. 8). This migration is expressed on the orthogonal plots as a roughly straight line that does not go to the origin (*L* of Fig. 8). The direction of magnetization removed along this straight line was calculated using either vector subtraction or LINEFIND. This magnetization is presumably carried by titanomagnetite euhedra with ilmenite subgrains visible in polished section. The toe (500–600°C, Fig. 6) in the thermal demagnetization curve and the 10% of magnetization left at 100 mT (Fig. 8) are probably carried by very fine grained magnetite disseminated throughout the groundmass. This tendency for the inclination to shallow, and even become negative, indicates that a small reversed overprint may be present, but it is so poorly preserved that its direction can't be described accurately. Such an overprint could be of Tertiary age, associated in time with burial and water circulation during deposition of the overlying Eureka Sound Formation, when reversals were frequent.

C-*type magnetization*

No coherent magnetic memory could be determined at 10 of the sites. These sites show the third type of behaviour (*C*-type magnetization). Specimens from seven sites (AXW16, AXB34, AXV08, AXV33, AXS03, AXS05, and AXC23) had directions of magnetization that are widely scattered directions initially and that behaved erratically upon demagnetization. Evidently, any original magnetizations that the rocks might have possessed have been reduced by viscous decay to their minimum or random level (Irving *et al.* 1961). This behaviour is presumably dictated by the generally low oxidation conditions under which it is expected that a proportion of the volcanic pile would have magnetic grains with large effective grain sizes and corresponding coercities too low to retain any memory of the Cretaceous geomagnetic field. Sites AXB36, AXB51, and AXS01 each had disparate directions corresponding to separate hand samples, suggesting possible field misorientation, which, in view of the difficulties in obtaining orientations under commonly miserable weather conditions, is not improbable.

The stability of *A*- and *B*-type magnetizations has been studied by observing the effect of heating to moderate temperature for several months. Pairs of specimens with closely matched magnetizations from the same hand sample were chosen. They were first demagnetized at 30 mT to remove the PEF magnetization. They were then heated to 250°C in the present Earth's geomagnetic field (vertical component, 0.05 mT) for increasing intervals and cooled to room temperature; then the vertical component of magnetization was measured (Fig. 9). This experiment was carried out to estimate the effect of deep burial and heating of the volcanics to lower temperatures (say 200°C) for a long period of time (tens of thousands of years). One specimen from each of the pairs was placed upright and one inverted in the furnace, so by taking sums and differences of the pairs the decay of the natural remanent magnetization (NRM freed of the PEF magnetization) and buildup of viscous remanent magnetization (VRM) can be monitored (Pullaiah *et al.* 1975). Both *A*-type and *B*-type specimens show no decay of remanence after 7 months at 250°C (Fig. 9). Stratigraphic reconstructions indicate that the maximum burial depth of Isachsen volcanic rocks was achieved in the Paleocene and did not exceed 2500 m in the Bunde Fiord area (localities AXW, AXB, AXV, Fig. 3) (Embry and Osadetz 1988). Maximum burial depth of the Strand Fiord Formation at Strand Fiord (AXS, Fig. 3) in the Eocene probably did not exceed 4500 m depth (Ricketts 1986;

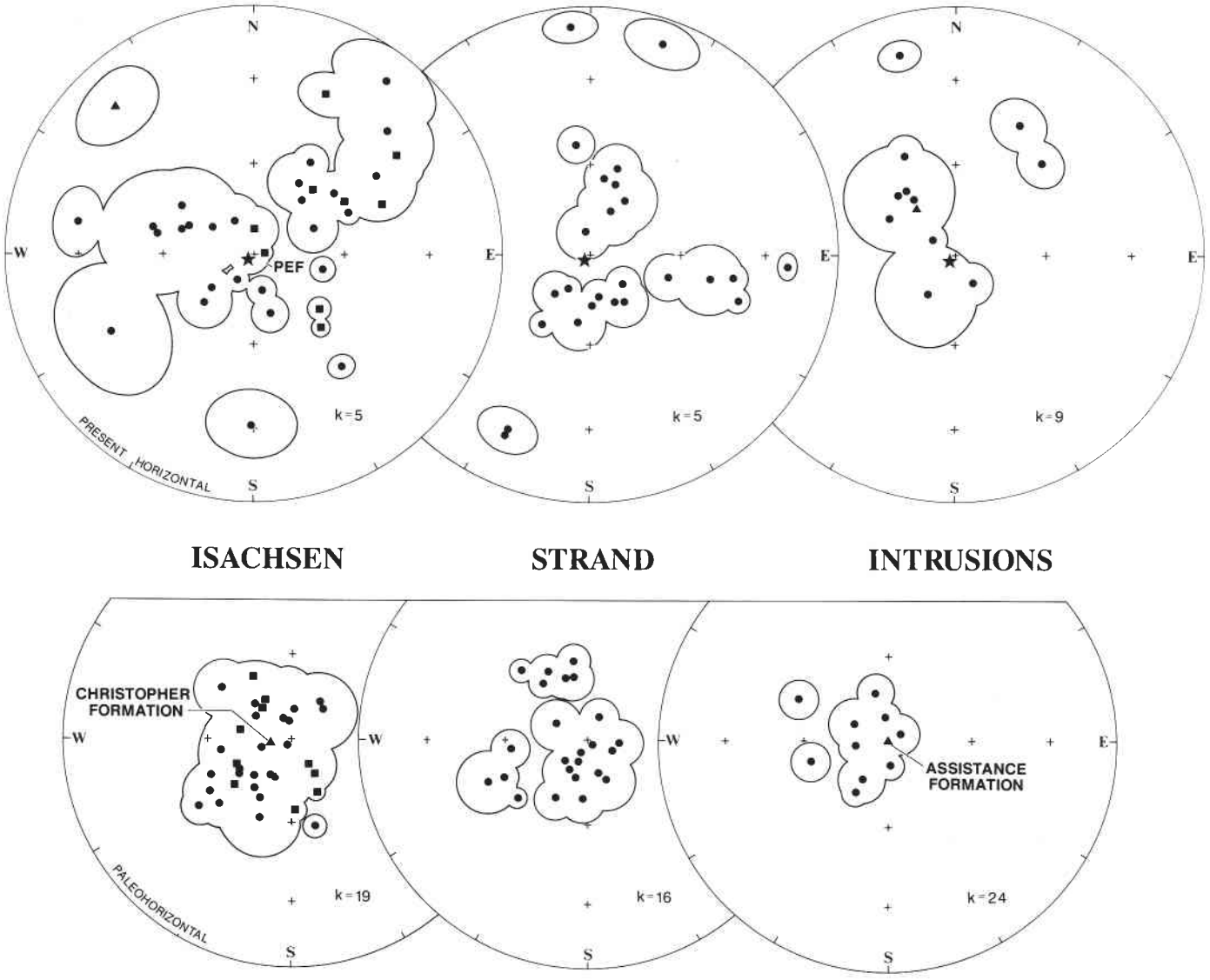


FIG. 10. Site-mean directions for three rock units, *Y* data set. Solid dots are normal directions; solid squares are sites with reversed directions that have been inverted. Sites with $\alpha_{95} > 20°$ or $k < 10$ are listed in Table 2 but are not plotted. Envelopes of error circles (α_{95}) are plotted. Paleohorizontal for the intrusions is estimated from the bedding attitude of the enclosing sediments. Directions with respect to present horizontal above, paleohorizontal below; k is Fisher's precision; and the star marks the direction of the present Earth's field in the sample area.

Embry and Osadetz 1988). Even if we assume an elevated geothermal gradient (say 40 mK/m), neither formation is expected to have reached temperatures in excess of 200°C. It is therefore very unlikely that the principal component of the *A*- and *B*-type magnetizations has been thermally overprinted during the course of its geological history.

In this heating experiment, there was negligible buildup of VRM in the *A*-type specimen, but in the specimen of the much more common *B*-type a large thermally enhanced VRM was developed. Clearly the development of a strong VRM during deep burial by a volcanic pile is a potent mechanism for producing large, positive magnetic anomalies. The Alpha Ridge (Fig. 1) was formed during the Cretaceous and may be consanguineous with Strand Fiord volcanism. If this is so and if it is composed, like much of the on-land volcanics, of rocks with *B*-type magnetization, then at crustal depths where the temperatures may be as much as 200°C, large VRM's would be produced. Such VRM, together with the strong normally directed remanent magnetization, would produce intensities of about $3-4$ A m^{-1}, a value that compares well with those assumed ($2.8-5$ A m^{-1}) by Taylor (1983) to model the magnetic anomalies of the Alpha Ridge.

Restoration of beds to paleohorizontal

The presence of plunging structures at sample localities has been considered because incorrect restoration could yield spurious declinations (MacDonald 1980). Throughout most of the sampled area, the plunge of folded structures is small. This is evidenced by the consistent exposure of single formations, several hundred metres thick, in the hinge of folds, several kilometres long. Elsewhere the plunge of large folded structures can be estimated using the distribution of poles to bedding. At the localities sampled, structures useful for estimating the plunge of major folds, such as small-scale parasitic folds or bedding–cleavage intersections, are absent.

At two localities (AXC01-04 and AXA01-06, Fig. 3), plunge cannot be estimated, but since dips are less than 30°,

TABLE 2. Site-mean directions

Site	*H*, *N* (*S*)	*M* (A m^{-1})	*D* (°)	*I* (°)	*k*	α_{95} (°)
EUR02	*7* (14)	0.67	214	69	143	5
EUR20	5 (*10*)	5.07	352	81	57	6
RUE41	*5* (9)	2.30	305	59	61	10
RUE42	*3* (6)		305	78	16	32
	4 (8)		028	−57	40	15
	7 (14)	2.13	224	71	11	20*
RUE43	*5* (8)	4.47	220	76	20	21*
RUE44	*6* (11)	2.30	201	87	42	10
RUE45	3, *6* (10)	1.27	046	75	26	13
AXW02	3, 6 (*11*)	0.60	297	76	30	9
AXW01	3, 5 (*9*)	1.37	296	54	61	7
AXW03	3, 7 (*14*)	9.16	216	74	28	8
AXW04	3, 7 (*11*)	0.06	135	−75	31	8
AXW08	2, 3 (*6*)	1.15	236	55	45	10
AXW09	1, 1 (*4*)	5.69	234	48	236	6
AXW08−AXW09	3, 4 (*10*)	2.97	235	52	64	6
AXW10	3, 6 (*9*)	0.27	086	−67	42	9*
AXW11	4, 8 (*14*)	3.37	209	75	8	15*
AXW12	2, 4 (*7*)	3.33	202	59	19	14
AXW13, AXW14	2, 3 (*5*)	0.56	311	72	25	16
AXW18	3, 7 (*12*)	3.09	206	81	34	8
AXB44	3, 7 (*14*)	2.72	257	61	31	7
AXB45	6 (*12*)	14.8	338	83	6	19*
AXB46	7 (*12*)	3.70	300	75	20	10
AXB47	2 (*3*)	3.92	329	64	17	31*
AXB50	3, 8 (*12*)	0.81	257	82	13	13
AXB49	8 (*13*)	3.99	202	75	38	7
AXB32	8 (*16*)	4.92	254	79	19	9
AXB33	4 (*6*)	3.02	255	71	29	13
AXB31	9 (*18*)	3.29	260	64	41	6
AXB31−AXB33	13 (*24*)	3.23	259	66	37	5
AXB43	9 (*16*)	7.16	213	85	40	6
AXB42	7 (*13*)	1.59	145	88	29	8
AXB41	6 (*10*)	6.91	160	78	21	11
AXB40	8 (*16*)	1.43	300	77	45	6
AXB39	2, 4 (*7*)	3.21	146	62	8	22*
AXB38	3, 9 (*14*)	4.08	115	81	44	6
AXB37	3, 7 (*11*)	2.55	154	75	18	11
AXB35	3, 8 (*12*)	0.87	185	70	26	9
AXB01	4, 8 (*15*)	1.08	358	−64	59	5
AXB02	4, 9 (*13*)	0.91	330	−79	20	10
AXV10	3, 6 (*12*)	0.34	065	−68	223	3
AXV01	3, 7 (*13*)	0.46	335	−69	165	3
AXV02	3, 9 (*16*)	0.63	328	−75	91	4
AXV03	4, 9 (*13*)	0.18	109	−71	29	8
AXV04	3, 6 (*12*)	2.22	165	58	116	4
AXV05	4, 9 (*16*)	3.04	208	66	37	6
AXV09	3, 9 (*15*)	3.65	236	67	51	5
AXV30	3, 9 (*17*)	0.03	147	−64	56	5
AXV31	3, 8 (*11*)	0.73	142	−74	55	6
AXV32	4, 9 (*16*)	0.54	051	−63	87	4
AXV34	3, 9 (*16*)	4.36	218	68	91	4
AXV35	3, 9 (*14*)	6.01	239	68	37	7
AXV36	3, 9 (*17*)	2.82	228	55	21	8
AXV37	2, 5 (*7*)	1.65	246	58	123	6
AXA05	3, 9 (*17*)	3.52	036	74	102	4
AXA06	3, 9 (*16*)	1.18	002	80	44	6
AXA01	3, 8 (*9*)	7.54	344	84	38	9
AXA02	3, 9 (*18*)	1.25	206	88	66	4
AXA03	3, 9 (*16*)	3.06	024	81	29	7
AXA04	3, 9 (*19*)	3.24	097	79	16	9
AXS04	3, 5 (*9*)	15.0	214	77	34	9
AXS02	3, 6 (*12*)	3.18	211	66	45	7
AXS07	2, 4 (*5*)	2.98	203	76	57	10
AXS09	3, 4 (*5*)	5.65	264	62	110	7
AXS15	3, 9 (*17*)	3.62	316	55	96	4
AXS14	3, 7 (*10*)	3.64	321	64	79	6
AXS13	3, 4 (*8*)	4.08	349	62	77	6
AXS12	3, 4 (*8*)	1.65	328	62	86	6
AXS11	2, 5 (*10*)	0.34	338	67	46	7
AXS10	2, 3 (*5*)	4.28	345	67	90	8
AXS10−AXS11	4, 8 (*15*)	2.82	340	67	57	5
AXS25	3, 4 (*8*)	4.14	228	78	180	4
AXS24	4, 6 (*10*)	2.10	248	51	21	11
AXS22	3, 5 (*10*)	3.97	231	58	144	4
AXS21	3, 4 (*7*)	3.29	246	57	127	5
AXC01	8 (*10*)	1.53	318	86	8	18*
AXC02	11 (*21*)	2.83	345	72	25	7
AXC12	3, 7 (*14*)	5.17	266	78	54	6
AXC13	3, 8 (*16*)	3.42	166	82	67	5
AXC14	3, 7 (*14*)	1.90	184	68	7	17*
AXC09	2, 4 (*8*)	0.01	113	73	21	12
AXC10	4, 8 (*14*)	0.08	051	85	35	7
AXC11	4, 8 (*12*)	0.09	288	85	20	10
AXC11	1, 2 (*4*)	0.05	108	−35	51	17
AXC04	3, 9 (*15*)	0.07	265	81	27	8
AXC04, AXC09, AXC10, AXC11	13, 29 (49)	0.07	270	90	20	5

NOTES: Sites ordered as in Table 1. *H*, *N* (*S*) are the number of hand samples, cores (specimens). The number in italics indicates whether core or specimen was accorded unit weight in the calculations. *M* is the average NRM intensity; *D*, *I* are the declination (east of geographic north), inclination of the site-mean direction relative to paleohorizontal; *k* is the within-site precision (Fisher 1953); and α_{95} is the radius of the circle of confidence ($P = 0.05$). In the analysis of inclination, sites denoted by an asterisk with $k < 10$ or $\alpha_{95} > 20°$ were excluded (the remainder constitute the *Y* data). Also, stratigraphically adjacent site pairs AXW08–AXW09, AXB31–AXB33, and AXS10–AXS11 have individual means that are not significantly different ($P = 0.05$), and the means of the combined specimens from each pair were used. Sites AXW13 and AXW14 (Table 1) were averaged together and treated as one site (AXW13) because each had only one surviving hand sample. Site *EUR20* is from Wynne *et al.* (1983). Procedures used to isolate the magnetization present are as follows: *EUR02*, directions calculated from slope of line obtained from AF demagnetization of 11 specimens in the range 20–80 mT (11 specimens), cleaned at 550°C (3 specimens); *EUR20*, 30 mT (9), 550°C (1); *RUE41*, 60–100 mT (5), 600°C (4); *RUE42*, 100 mT (3), 60–100 mT (4), 575°C (2), 600°C (5); *RUE43*, 30–100 mT (1), 70–100 mT (6), 300–500°C (1); *RUE44*, 40–60 mT (4), 30–70 mT (5), 400–600°C (2); *RUE45*, 40–100 mT (4), 70–90 mT (4), 400–575°C (2); *AXW02*, 10–20 mT (1), 50–60 mT (1), 200–500°C (9); *AXW01*, 10–20 mT (4), 20–90 mT (1), 100–400°C (1), 400–550°C (3); *AXW03*, 10–50 mT (13), 200–700°C (1); *AXW04*, 30–90 mT (10); 100–575°C (1); *AXW08*, 10–30 mT (4), 300–700°C (1), 100–500°C (1); *AXW09*, 10–70 mT (1), 100–600°C (3); *AXW10*, 10–40 mT (7), 400–600°C (2); *AXW11*, 10–40 mT (7), 200–525°C (6), 400–575°C (1); *AXW12*, 15–50 mT (4), 20–90 mT (1), 200–500°C (2); *AXW13*, *AXW14*, 20–100 mT (3), 400–650°C (2); *AXW18*, 50–100 mT (10), 20–100 mT (1), 100–550°C (1); *AXB44*, 10–30 mT (12), 10–50 mT (1), 300–525°C (1); *AXB45*, 30 mT (2), 400°C (3), 400–550°C (7); *AXB46*, 30 mT (1), 300°C (8), 200–550°C (3); *AXB47*, 30 mT (1), 50 mT (1), 100–400°C (1); *AXB50*, 20–60 mT (1), 300–600°C (11); *AXB49*, 20–80 mT (1), 300°C (12); *AXB32*, 30 mT (15), 400°C (1); *AXB33*, 30 mT (5), 450°C (1); *AXB31*, 20–40 mT (16), 300–550°C (2); *AXB43*, 20–60 mT (1), 300°C (15); *AXB42*, 40 mT (2), 20–80 mT (5), 300–550°C (6); *AXB41*, 30–40 mT (8), 300–550°C (2); *AXB40*, 30 mT (2) 20–40 mT (13), 550°C (1); *AXB39*, 30 mT (6), 550°C (1); *AXB38*, 40 mT (13), 550°C (1); *AXB37*, 30 mT (6), 20–40 mT (4), 550°C (1); *AXB35*, 20 mT (11), 300°C (1); *AXB01*, 50 mT (13), 500°C (2); *AXB02*, 30 mT (8), 20–30 mT (3), 400–550°C (2); *AXV10*, 20–80 mT (11), 100–525°C (1); *AXV01*, 10–50 mT (12), 400–600°C (1); *AXV02*, 20–60 mT (11), 40 mT (4), 550–650°C (1); *AXV03*, 10–90 mT (12), 100–600°C (1); *AXV04*, 10–60 mT (11), 100–600°C (1); *AXV05*, 10–100 mT (1), 100–500°C (10); *AXV09*, 20–60 mT (14), 100–650°C (1); *AXV30*, 20–60 mT (16), 100–575°C (1); *AXV31*, 20–80 mT (10), 100–500°C (1); *AXV32*, 20–80 mT (15), 100–600°C (1); *AXV34*, 10–30 mT (15), 300–650°C (1); *AXV35*, 10–30 mT (13), 100–600°C (1); *AXV36*, 20–60 mT (1), 10–30 mT (1), 100–500°C (15); *AXV37*, 10–50 mT (1), 200–500°C (6); *AXA05*, 10–90 mT (1), 300–500°C (16); *AXA06*, 10–20 mT (1), 300–500°C (15); *AXA01*, 20–50 mT (2), 300–500°C (7); *AXA02*, 10–30 mT (4), 150–575°C (14); *AXA03*, 10–30 mT (2), 150–550°C (14); *AXA04*, 10–30 mT (1), 10–90 mT (1), 300–500°C (17); *AXS04*, 10–30 mT (7), 30–90 mT (1), 200–500°C (1); *AXS02*, 20–50 mT (10), 20–100 mT (1) 200–500°C (1); *AXS07*, 20–50 mT (4), 200–500°C (1); *AXS09*, 20–80 mT (4), 300–550°C (1); *AXS05*, 20–80 mT (16), 100–600°C (1); *AXS14*, 20–50 mT (7), 80 mT (1), 10–100 mT (1), 100–525°C (1); *AXS13*, 10–30 mT (7), 300–500°C (1); *AXS12*, 10–40 mT (6), 40–100 mT (1), 525–700°C (1); *AXS11*, 10–30 mT (8), 100–550°C (2); *AXS10*, 20–80 mT (3), 100–650°C (2); *AXS25*, 20–50 mT (6), 100–550°C (2); *AXS24*, 20–50 mT (9), 100–550°C (1); *AXS22*, 20–50 mT (8), 10–100 mT (1), 100–575°C (1); *AXS21*, 20–40 mT (5), 10–60 mT (1), 100–525°C (1); *AXC01*, 20 mT (5), 500°C (5); *AXC02, 03*, 30 mT (14), 40 mT (6), 500°C (1); *AXC12*, 30 mT (7), 40 mT (5), 500°C (2); *AXC13*, 30 mT (8), 40 mT (6), 500°C (2); *AXC14*, 30 mT (7), 40 mT (6), 500°C (1); *AXC09*, 50 mT (4), 75 mT (1), 100 mT (2), 300°C (1); *AXC10*, 50 mT (8), 100 mT (5), 300°C (1); *AXC11*, 50 mT (7), 100 mT (4), 300°C (1); *AXC04*, 30–50 mT (9), 100–300°C (1).

the effect of any possible plunge on the declinations is less than 5°. Data from these localities therefore are used without plunge correction.

For two folded structures, plunge of geological structure has been estimated from poles to bedding. The first is the structure on which localities AXB31-51, AXV30-37, AXB01-02, and AXV01-10 are located (Fig. 3). The axis of this fold, calculated using 29 poles to bedding, dips 16° at 250°E approximately. Fischer (1985) interpreted this structure as a simple continuous footwall syncline plunging beneath the Bunde Fiord thrust fault. The second structure is the Muskox Ridge diapiric anticline, north of Strand Fiord (AXS02-09, Fig. 3). The axis of this fold is estimated to dip 3° at 110°E approximately. In both structures, the effect of plunge correction on the paleomagnetic directions is less than 5°, and therefore no corrections are applied.

That localities AXS21-25 and AXW01-18 are situated on structures with little or no plunge is indicated by the persistent occurrence, over several kilometres, of one formation in the hinge of the fold. Locality AXS10-15 is in a basinal structure in which the plunge of the folds is equal to and in the same direction as the bedding dip, so the beds are correctly restored to paleohorizontal by a single rotation.

Summary of directions

Mean directions of the principal magnetization of sites and localities are listed in Tables 2 and 3 and plotted with respect to both present horizontal and paleohorizontal in Figs. 10 and 11. They are arranged in three groups: Isachsen Formation, Strand Fiord Formation, and intrusions. Data from site AXB50 in the Christopher Formation (Albian) is included with the first group (Fig. 10). Four sites (AXC04, AXC10, AXC11, AXC12) from the sedimentary Permian Assistance Formation lie below a large sill and, with the exception of one hand sample from site AXC11, have directions like those of the sill (Table 2), not the shallower reversed directions expected for Permian rocks (Wynne *et al.* 1983). The unblocking temperatures are mainly between 200 and 350°C and are always less than 400°C, and their magnetization was probably acquired as a result of reheating by the adjacent Cretaceous sill. They have been averaged together and plotted along with the intrusions (Table 2, Fig. 10).

Site directions for all three groups are clustered better after correction from present horizontal to paleohorizontal (precision *k* increases significantly to 18), but the grouping is still not very good. This question will be considered later.

Age of magnetization

During the early stages of demagnetization a soft component is removed, causing a systematic migration away from the PEF. An example is shown in Fig. 12. Evidently, the principal magnetizations were not produced recently.

The mean directions obtained from sites AXS02 and AXS04, which lie on the north limb of an anticline, and those from sites AXS07 and AXS09, which lie on the south limb, are widely divergent when plotted with the limbs in their present folded attitudes. When the directions are unfolded and plotted with respect to paleohorizontal they are in good agreement (Fig. 13). As already noted, clustering of site means on a regional scale is significantly improved after correction to paleohorizontal (Fig. 10). Hence the principal magnetization is older than Eurekan deformation and was acquired prior to the Middle Eocene.

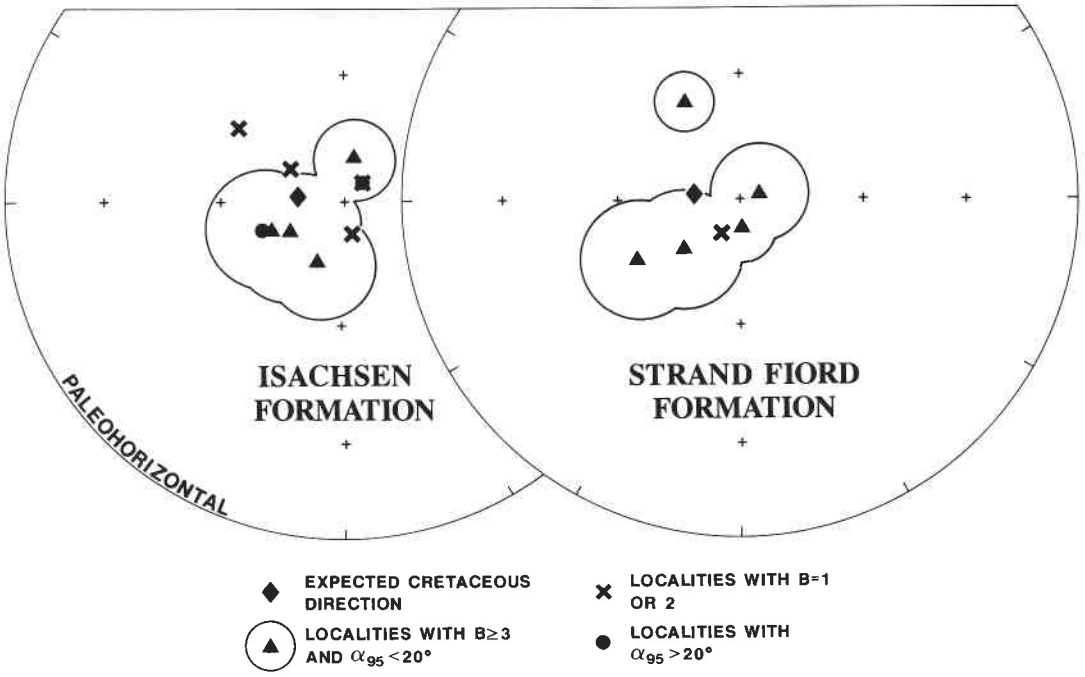


FIG. 11. Locality-mean directions for the Isachsen and Strand Fiord formations. Triangles are those localities that are well defined and are shown with corresponding errors (α_{95}); B is the number of sites.

TABLE 3. Rock-unit averages by locality

Locality	Rock unit	B (S)	M (A m^{-1})	D (°)	I (°)	k	α_{95} (°)	α_{63}/cos I (°)
Keybolte Peninsula	Porphyry dyke	1 (14)	0.67	214	69	143	5	5
Van Royen Ridge	Diabase sill	1 (10)	5.07	352	81	57	6	22
Mokka Fiord	Isachsen basalt equivalent	1 (9)	2.30	305	59	61	10	7
Blue Mountain	Isachsen basalt equivalent	2 (21)	1.78	052	84	42	40	84
Bunde Fiord (west)	Diabase sills	3 (34)	3.71	279	72	18	30	36*
	Isachsen basalt	4 (33)	1.73	248	71	12	27	36*
	Strand Fiord basalt	1 (12)	0.56	206	81	34	8	26
Celluloid Creek	Diabase dyke	1 (14)	2.72	257	61	31	7	8
(north section)	Isachsen basalt	1 (12)	3.70	300	75	20	10	20
	Christopher tuff	1 (12)	0.81	257	82	13	13	47
(south section)	Isachsen basalt	3 (40)	4.05	241	75	55	17	24*
	Strand Fiord basalt	7 (99)	3.51	175	83	55	8	34*
Bals Fiord	Isachsen basalt	3 (42)	4.08	015	80	142	10	23*
	Strand Fiord	3 (53)	2.52	072	85	105	12	53*
Strand Fiord								
"Anticline"	Strand Fiord basalt	4 (31)	6.69	228	72	44	14	20*
Bastion Ridge	Strand Fiord basalt	5 (58)	3.16	330	63	108	7	8*
"South of Camp"	Strand Fiord basalt	4 (35)	3.38	240	61	42	14	13*
Camp Five Creek								
"Pit stop"	Isachsen basalt	2 (28)	1.00	350	−72	45	38	28
South section (the creek)	Isachsen basalt	7 (97)	1.51	204	74	21	14	24*
North section (at the fiord)	Isachsen basalt	7 (98)	2.31	251	69	20	14	19*
Cape Stallworthy	Diabase sill	1 (21)	2.83	345	72	25	7	11
	Diabase sill	2 (30)	4.30	229	83	55	34	63
	Assistance sandstone	1 (49)	0.07	270	90	20	5	—

NOTES: B (S) are the number of sites (specimens) with $\alpha_{95} < 20°$ and $k > 10$ included in the average. Unit weight given to sites except when $B = 1$, and unit weight then given to specimen. Stratigraphically adjacent sites with mean directions not significantly different ($P = 0.05$) were treated as one site. M is the average NRM intensity; D, I are the declination east of geographic north, inclination of the mean direction relative to paleohorizontal. Mean directions for the Isachsen and Strand Fiord formations are shown in Fig. 10. An asterisk indicates those entries with $B \geq 3$; these are included in Fig. 17.

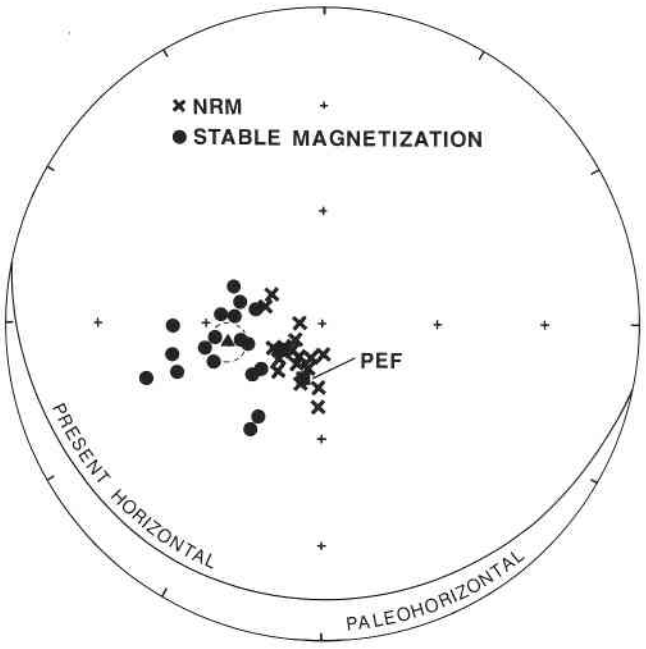


FIG. 12. Isachsen Formation, site AXB31. NRM directions cluster close to present Earth's field direction, but after cleaning (AF) the magnetization directions move away from the PEF. The mean magnetization direction (triangle) with α_{95} error circle (dashed) is shown.

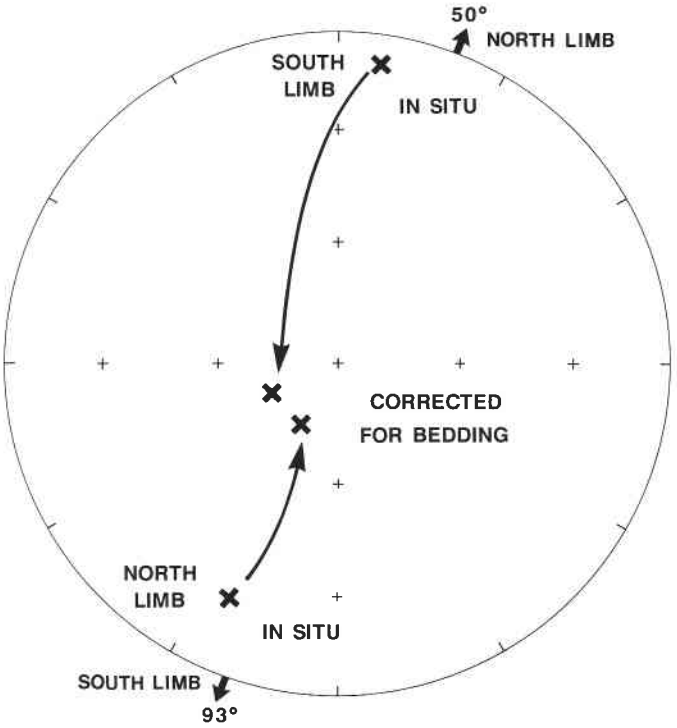


FIG. 13. *In situ* mean directions of the two "Anticline" localities come into much better agreement after correction for bedding attitude. In the analyses (with respect to paleohorizontal) the two localities were averaged and treated as one.

As described below, the reversal pattern is consistent with the global reversal time scale and indicates that the magnetizations originated at or close to the time the rocks were formed. In laboratory tests the remanence is resistant to long-term heating at temperatures in excess of those experienced by the rock during burial (Fig. 9). The rocks are fresh, many having their original delicate cooling textures intact, and there is no evidence of gross secondary alteration such as might produce magnetic overprinting. Therefore, we conclude that the principal magnetization of the Isachsen and Strand Fiord formations and coeval intrusions was acquired as the rocks cooled shortly after their formation in the Cretaceous.

Reversals of magnetization

Flows of the Strand Fiord Formation, the intrusions, and the tuffaceous sandstone from Christopher Formation are normally magnetized. The Srand Fiord flows are probably Albian to Turonian in age, based largely on their stratigraphic position (Embry and Osadetz 1988). Eleven of the 28 flows studied from the Isachsen Formation are reversely magnetized. The reversed flows occur at two horizons: in the lower part of the Isachsen Formation and in the upper third. The uppermost Isachsen flows are normally magnetized (Table 2; Fig. 14). All of the flows that we sampled in the Isachsen Formation, which ranges in age from Valanginian to Aptian (Wall 1983), lie in the upper portion of the Walker Island Member, which is late Barremian to Aptian (Embry and Osadetz 1988), or in equivalent strata. These biostratigraphic age assignments indicate that the normal magnetizations of the Strand Fiord Formation, the Christopher Formation, and the top of the Isachsen Formation were acquired during the Cretaceous normal polarity superchron. Hence we tentatively suggest that the uppermost horizon of reversed flows of the Isachsen Formation (sites AXB01, AXB02, AXW10, Fig. 14) corresponds to M_0, the youngest of the M sequence of reversed anomalies, and that the lower horizon of reversed flows (AXV10, AXV01, AXV02, AXV03, AXV30, AXV31, AXV32, AXW04, Fig. 14) correspond to the M_1 reversed anomaly.

The recognition of this reversal stratigraphy has proven to be helpful regionally as a correlation tool but potentially is of much more general importance because these Isachsen sections could provide the standard for dating accurately the base of the Cretaceous normal polarity superchron by both paleontological and isotopic methods. Further paleomagnetic and biostratigraphic work, particularly on microfossils and isotopic dating of the lavas, will be required to check this possibility. On biostratigraphic evidence, it seems generally to be agreed that the base of the M_0 reversed chron marks or corresponds to the Aptian–Barremian boundary, which has been assigned, by interpolation, an age ranging from 112 to 119 Ma (Harland *et al.* 1982; Kennedy and Odin 1982; Hallam *et al.* 1985; Lowrie and Ogg 1985/86). This difference of about 7 Ma translates to an uncertainty of approximately 30% in Early Cretaceous oceanic spreading rates, an important factor in tectonic reconstructions. Isotopic dating should also provide improved age control for the Aptian-Barremian boundary.

Analysis

When analysing directions of magnetization, there are two ways to proceed: by Fisher's statistics or by separate treatment of declinations and inclinations. If it is assumed that there have been no relative rotations amongst localities, then vector averages and paleopoles could be calculated directly. This is referred to as *method I*. However, it is possible that relative

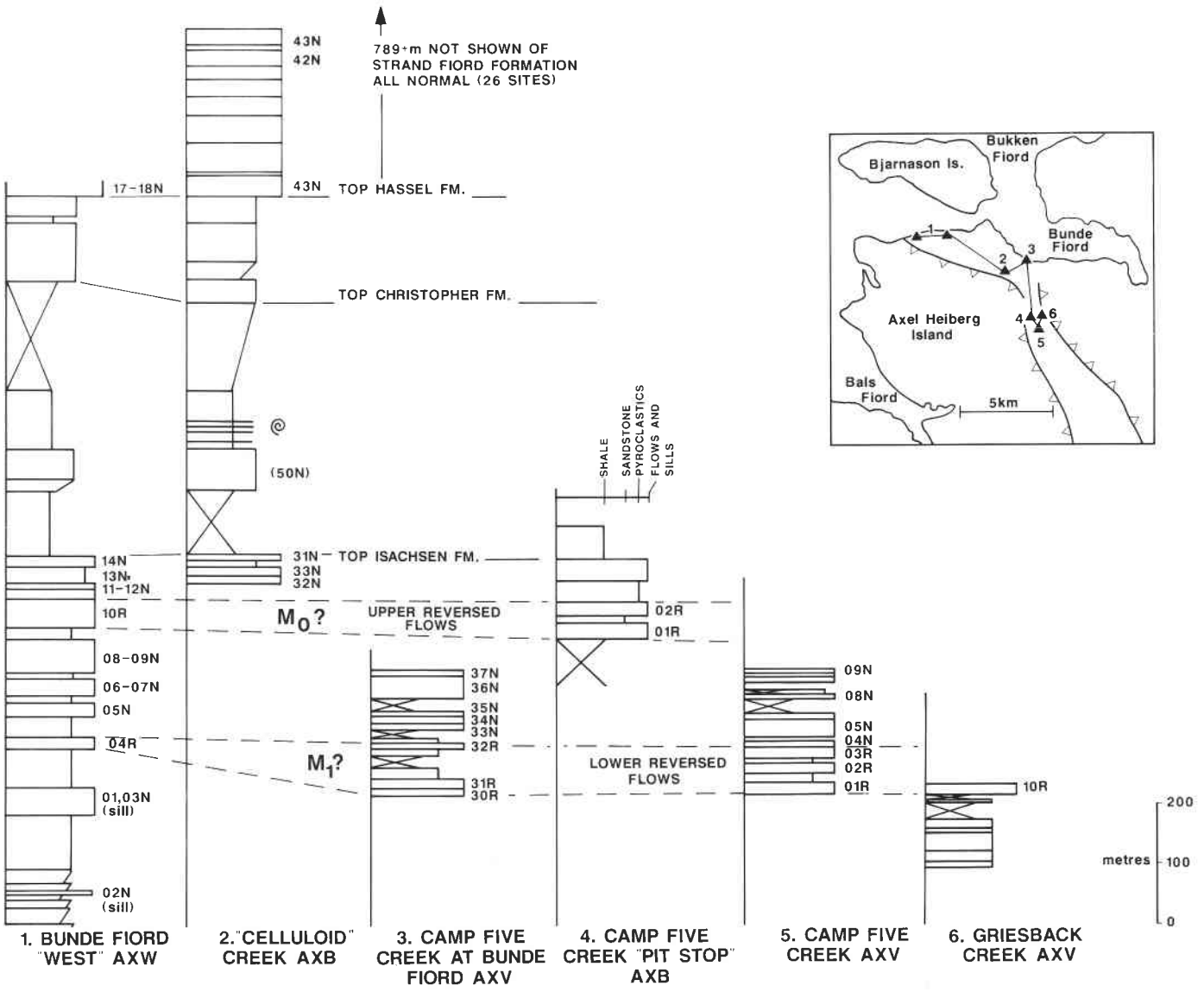


FIG. 14. Polarity sequence in sections from Bunde Fiord. N (R) denotes normally (reversely) magnetized flows. Site 50N and the 26 Strand Fiord Formation sites were sampled at equivalent stratigraphic levels in sections other than the Celluloid Creek section. The ammonite found in the Celluloid Creek section is *Beudanticeras* (*Grantziceras*) *glabrum* (Whiteaves) (J. A. Jeletzky, personal communication, 1984).

rotations have occurred, which means that declinations and inclinations must be analysed separately. This procedure is referred to as *method II*.

Data from lavas have been arranged into two sets: set *X*, which includes all sites, and set *Y*, from which less accurate or redundant data have been excluded. The latter have been identified as sites with α_{95} greater than 20° or with precisions (k) less than 10. Stratigraphically adjacent sites, with mean directions that were not significantly different ($P = 0.05$), were considered as having recorded the paleofield only for a single interval of time and were averaged together. It would have been desirable to eliminate possible duplication of directions produced as a result of inadvertent sampling of the same flow in a different section, but this could not be done because it is not possible to correlate flow-by-low among sections.

Method I—no relative rotations amongst localities

Two groups of localities represent multiple samplings of continuous folded structures. In west-central Axel Heiberg Island, localities AXS02-09 and AXS21-25 are situated within a folded yet continuous structural block. Locality AXS10-15 is separated from them by a fault, but the stratigraphic relationships do not suggest any large relative rotations. In northwestern Axel Heiberg Island, localities AXB31-51, AXB01-02, AXV30-37, and AXV01-10 are situated within a simple footwall syncline, which plunges shallowly beneath the Bunde Fiord thrust fault. The apparent simplicity of the geological structure suggests that these sites also are part of a coherent structural block and have not undergone relative rotations. These structural interpretations are consistent with the assumption made in method I. On this assumption, the dispersion of site-mean directions within and amongst localities is ascribed to the effects of paleosecular variation plus uncertainties in the paleomagnetic data.

The average of the *Y* data set ($B = 55$ sites) is listed in Table 4. It shows a $33 \pm 24°$ ($P = 0.05$) counterclockwise rotation (Fig. 15) but no significant paleolatitudinal displacement ($-1 \pm 5°$, Fig. 16) relative to the cratonic reference paleo-

TABLE 4. Summary by method I

	B (b)	Lat. (°N)	Long. (°W)	D (°)	I (°)	k	α_{95} (°)	α_{63} (°)	Lat. (°N)	Long. (°E)
Select flows' mean	55	80.0	92.0	247.7	79.7	18	4.7	2.7	64.6	220.7
Mean of poles	55					7	8.1	4.6	66.1	220.8
Cratonic paleopole	(4)						5.4	2.6	68.9	193.4

NOTES: B is number of site means; b is the number of studies used. The two-tier analysis of Watson and Irving (1957) yields a mean within-site precision of $k_W = 36$ and between-site precision of $k_B = 20$.

pole (Table 4). The large error of 24° (calculated using the method of Demarest (1983)) in the rotation estimate is caused partly by the steep inclinations and partly by the fact that the dispersion of site-mean directions ($k = 18$) and corresponding poles ($k = 7$) is substantial (Fig. 10). The rotation, however, is significant and similar in magnitude to that observed in the Permian Esayoo Formation of Ellesmere Island (counterclockwise 36 ± 8° (Wynne *et al.* 1983)). These results may be explained by supposing that the Esayoo localities on Ellesmere Island and the Strand Fiord – Isachsen localities on Axel Heiberg Island were contained within a single large tectonic domain (20 000 km^2) that has rotated approximately as a whole relative to North America.

Method II—relative rotations amongst localities permitted

As already noted, the dispersion, although much improved after correction for tilting, remains higher than expected for paleosecular variation. The angular standard deviation of site poles (Y data set) is 32°, much larger than the 22° obtained from the geomagnetic field in the past 5 Ma at latitudes of 70° (MacFadden and McElhinny 1984). The value 32° also may be compared with 14° recently obtained from essentially contemporaneous mid-Cretaceous alkali intrusions of Arkansas (Globerman and Irving 1988), which translates into an expected 20° at latitude 70°, again much less than that observed in the present study. Relative rotation amongst sampling localities could be responsible for this enhancement of dispersion, and this possibility is now considered. Hence, by method II, declinations and inclinations are analysed separately. Declinations are considered first.

Mean declinations grouped by locality are shown in Fig. 17. The data are arranged so that the north direction is the expected declination, based on the cratonic reference paleopole. The Permian Esayoo determination is included for comparison. In order to give greater visual weight to those sites in which the declination is defined better, we made the lengths of the declination lines proportional to the cosine of the locality mean inclinations; as inclinations become steeper, declination anomalies are more difficult to detect, and the lines are correspondingly shorter. Seven rock units have declinations that are counterclockwise, three are clockwise, and one shows no relative rotation (Fig. 17).

We do not claim that each of the individual *relative* rotations (between-location) shown in Fig. 17 is real. Indeed, with one exception (8), the error limits show that they are not. Relative rotations amongst sampling localities are permitted by our data but are not required by them. The apparent differences observed amongst locality-mean declinations could, equally well, be a measure of the error in determining declination when, as in the lavas, inclination is steep. As previously discussed, the apparent continuity of structures makes large relative rotations improbable on geological grounds. However, this does not, we suggest, overshadow the main observation—that the declinations tend systematically to be less than those expected—and that this tendency indicates an overall counterclockwise rotation. Hence the analysis of declinations separately (method II) is consistent with the results obtained by method I.

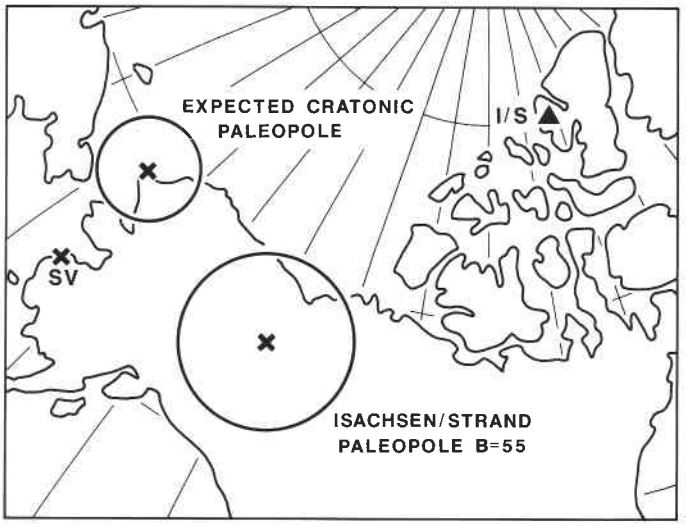


FIG. 15. Paleopole (assuming no rotations amongst localities) obtained by Method I compared with the mid-Cretaceous cratonic reference paleopole for North America (Table 4). B is the number of sites used in the average. SV is the paleopole from Svalbard dolerites (Vincenz and Jelenska 1985). Standard error circles ($P = 0.37$) are shown. I/S marks the Isachsen – Strand Fiord sample localities.

Inclinations are considered next. The data of Table 2 were averaged in two different ways: by a modified Fisherian procedure (MF) and by the procedure of Cox and Gordon (1984) (CG). The X and Y data sets were analysed in both ways (Table 5). Firstly (MF procedure), the mean inclination was obtained by first rotating all site-mean directions about a vertical axis so that the mean declination of each locality was equal to zero. One grand Fisher average was then calculated. This procedure artificially reduces dispersion and yields errors that are probably a little too small. Averages for each rock unit are given in Table 5. Secondly (CG method, which was designed originally to estimate paleolatitude from data obtained from vertical drill core), inclinations were first converted into paleocolatitudes, and their mean was calculated. A small negative correction (based on a certain model of paleosecular variation) was then made in order to arrive at an estimate of the true mean paleocolatitude. Mean paleolatitude was then calculated from the latter.

Errors (quoted as standard errors) obtained by either procedure are in approximate agreement, but as expected, those by the CG method tend to be a little greater and are the more realistic (Table 5). Because it is not possible to correct for between-locality redundancy, as noted above, the number of

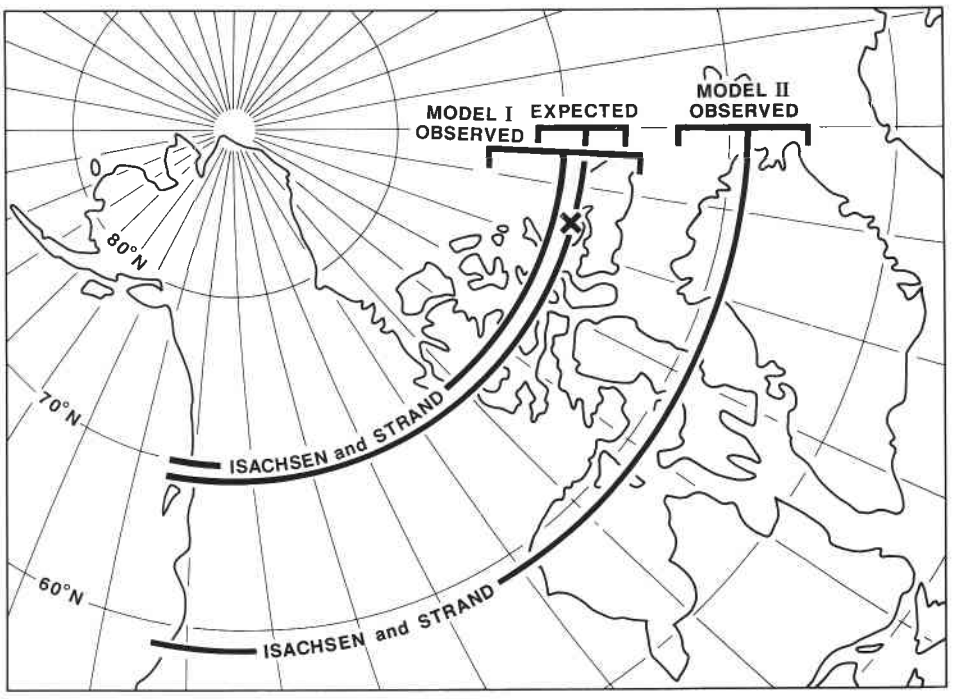


FIG. 16. Paleolatitudes for the Isachsen and Strand Fiord formations obtained by method I and method II compared with the expected for mid-Cretaceous cratonic North America. Standard error bars ($P = 0.37$) are shown. In method II the Y data set (Table 5) and the Cox and Gordon (1984) averaging procedure were used. The × marks the study area.

truly independent observations may be somewhat fewer than the number used in the calculations. It is therefore possible that our errors are too small but by no more than 10%.

These data may then be compared with the reference field for cratonic North America (Fig. 2). The mean inclinations (mean paleocolatitudes) of flows calculated by either method are shallower (greater) than those expected for Early Cretaceous rocks deposited at a paleolatitude of 69°N (Table 5). Although the difference between the observed and expected mean inclinations is only 5° (Table 5), the flattening tendency is clear and systematic across the collection and apparently is significant. In paleomagnetic work it is not generally accepted that such small differences are detectable. Nevertheless, because the data have been produced from extensive field collections and detailed laboratory analysis, we argue that given the assumption of method II, the discrepancy is real. The result obtained by applying the CG procedure to the Y data set is plotted in Fig. 16. The standard error bars ($P = 0.37$) only begin to overlap (that is, the means become no longer significantly different at $P = 0.05$) when Axel Heiberg and Ellesmere islands are moved away from the Early Cretaceous paleopole by roughly 300 km. This would place them snuggly against Greenland in pre-Tertiary reconstructions of this region (Fig. 18).

The difference between the inclinations calculated in method I and II are to be expected because in method I the calculation of a mean from directions with a wide range of declinations and steep inclinations could generate an overly steep mean inclination (79°). In method II the inclination-only analysis avoids this "wrap around problem," but in doing so it may underestimate the mean inclination. The inclination of the intrusions agrees with that expected (Table 5), but as already noted, "bedding" corrections are not very reliable, and our collection of intrusive samples is sporadic and by no means representative.

Discussion

Our data have consequences for Arctic tectonics. Both methods of analysis yield results that indicate that the counterclockwise rotation observed in the Permian Esayoo Formation on Ellesmere Island extends to the Cretaceous lavas of Axel Heiberg Island. As Greenland, rotating about a nearby Euler pole (−11°, 71.5°N, 126.3°W (Srivistava 1978)), moved away from Labrador in the south and towards the Queen Elizabeth Islands in the north a component of left-lateral shear would be imposed over the sampling region. The rotations we observe were presumably produced in response to this general stress field.

The mean inclinations and hence the paleolatitudes obtained by the two methods, however, are not in very good agreement. The steeper mean inclination obtained from method I corresponds to that predicted by the cratonic reference field, and there is no evidence of movement of the Queen Elizabeth Islands in a paleolatitudinal sense relative to cratonic North America. Although errors would permit motion of several hundred kilometres along the Northwest Passage (M'Clure Strait − Lancaster Sound lineament, Fig. 1), none is required by the data analysed according to method I. The result by method I is consistent with the existence of a large gore between Greenland and Ellesmere Island in the reconstructed Arctic (Figs. 2, 16). The shallower mean inclination obtained by method II allows Axel Heiberg and Ellesmere islands to be placed next to Greenland in the mid-Cretaceous, without appreciable opening of Nares Strait (Fig. 18). The absence of any longitudinal control in paleomagnetic studies precludes us from commenting on the amount of sinistral displacement of Ellesmere Island relative to Greenland. In order to achieve the reconstruction of Fig. 18, Axel Heiberg and Ellesmere islands (and presumably the rest of the Queen Elizabeth Islands) have been moved paleolatitudinally with respect to cratonic North America. Such motion would have had to occur either north

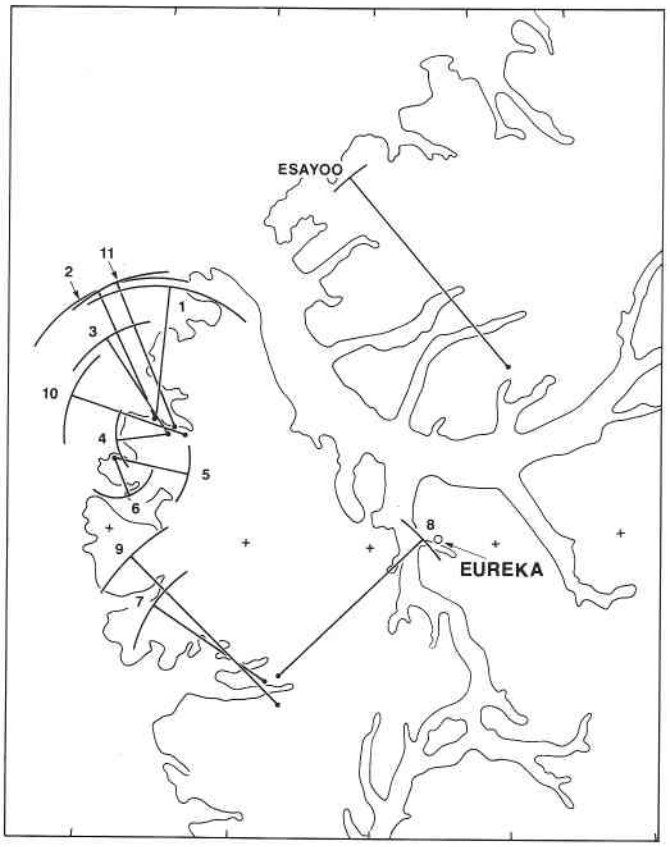


FIG. 17. Relative rotations, method II. The angle between the lines and geographic north is equal to the mean angle between the observed and expected direction (based on the expected cratonic paleopole) of the horizontal component of magnetization. The lines are proportional in length to cos I, where I is the mean inclination of the rock unit at that locality. The error fans are equal to $\pm \alpha_{63}$ /cos I as listed in Table 3 with the mean directions. (1) Diabase sills, Bunde Fiord west; (2) Isachsen basalts, Bunde Fiord west; (3) Isachsen basalts, south section, Celluloid Creek; (4) Strand Fiord basalts, south section, Celluloid Creek; (5) Isachsen basalts, Bals Fiord; (6) Strand Fiord basalts, Bals Fiord; (7) Strand Fiord basalts, "Anticline"; (8) Strand Fiord basalts, Bastion Ridge; (9) Strand Fiord basalts, "south of camp"; (10) Isachsen basalts, south section, Camp Five Creek; (11) Isachsen basalts, north section, Camp Five Creek at the fiord.

of or along the Northwest Passage, as Hamilton (1983) suggested. However, it must be emphasized that the uncertainties are such that the paleomagnetic evidence, as it stands at present, cannot discriminate between the reconstructions of Figures 16 and 18. Data from Cretaceous rocks of Ellef Rignes Island (Fig. 3) and from Permian rocks of Ellesmere Island (Wynne *et al.* 1983) are consistent with either possibility.

Paleomagnetically, the central question concerns the magnitude of the between-site dispersion calculated by method I. Is it really too large to truly reflect the magnitude of the Cretaceous secular variation? If it is (not) then method II (I) is the more applicable, and Fig. 18 (method I, Fig. 16) provides the correct interpretation of the inclination data. However, it should be noted that errors in site means are not insignificant (Table 2), because the intrinsic within-site dispersions are substantial. Errors in site means are estimated to correspond on average to a precision $k_W = 36$, using the two-tier analysis of Watson and Irving (1957), and are a consequence of the gen-

TABLE 5. Mean inclinations and colatitudes by method II

Rock unit	B	Inclination (°) MF	Inclination (°) CG	Paleocolatitude (°) MF	Paleocolatitude (°) CG
		X, ALL DATA			
Intrusions	15	78 ± 3	79^{+5}_{-4}	23^{+4}_{-6}	21^{+8}_{-9}
Isachsen	39	74 ± 3	72^{+3}_{-3}	30^{+4}_{-4}	32^{+4}_{-5}
Strand Fiord	26	73 ± 3	74^{+3}_{-3}	31^{+4}_{-4}	29^{+5}_{-5}
Isachsen + Strand Fiord	65	74 ± 2	73^{+2}_{-2}	30^{+3}_{-3}	31^{+4}_{-4}
All	80	75 ± 2	74^{+2}_{-2}	28^{+3}_{-3}	30^{+3}_{-4}
		Y, SELECTED DATA			
Intrusions	10	75 ± 4	77^{+5}_{-4}	28^{+7}_{-7}	25^{+7}_{-9}
Isachsen	31	74 ± 3	72^{+3}_{-3}	30^{+5}_{-5}	32^{+4}_{-5}
Strand Fiord	24	73 ± 3	75^{+3}_{-3}	31^{+4}_{-5}	28^{+5}_{-5}
Isachsen + Strand Fiord	55	74 ± 2	73^{+3}_{-2}	31^{+3}_{-3}	31^{+4}_{-4}
All	65	74 ± 2	74^{+2}_{-2}	30^{+3}_{-3}	30^{+3}_{-4}
Expected		79 ± 1		21 ± 3	

NOTES: B is the number of sites. MF is the modified Fisher procedure, CG the procedure of Cox and Gordon (1984). Standard errors quoted. CG inclination (I) is calculated from CG paleolatitude, and MF paleocolatitude (θ) is calculated from the MF inclination using the relationship tan (I) = 2 tan (λ) where λ, the paleolatitude, equals ($90 - \theta$). The expected inclination and colatitude are calculated using the cratonic reference paleopole of 68.9°N, 193.4°E, $A_{95} = 5.4°$, $A_{63} = 2.6°$ based on the results from four cratonic studies (Irving and Wynne 1988). X and Y are the data sets as described in the text. Results from Y data set by the CG procedure are converted to paleolatitudes and plotted in Fig. 16.

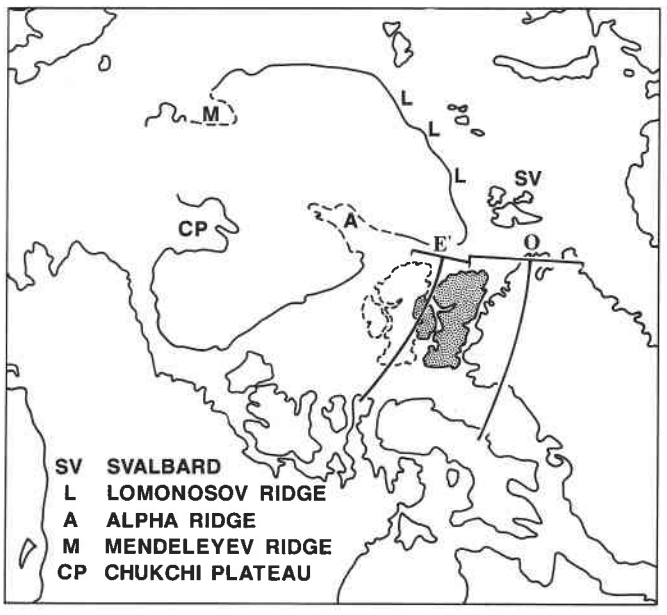


FIG. 18. Early Cretaceous reconstruction of Ellesmere and Axel Heiberg islands and Greenland, method II. The dashed outline shows the present-day position of Axel Heiberg and Ellesmere islands relative to Nort America. The stippled area shows a possible Early Cretaceous position that would bring the expected paleolatitude, E′, into agreement with the (method II) observed O paleolatitude (done here manually so their standard error bars overlap). The reconstruction was accomplished using two rotations: (1) 12.5° about 50.5°N, 110.0°E (Le Pichon *et al.* 1977) closes the Eurasian Basin and the North Atlantic; (2) −26.7° about 85°N, 129°E (Le Pichon *et al.* 1977) rotates Europe and Greenland (and corresponding 2000 m isobath) to North America, closing the Labrador Sea and Baffin Bay.

erally moderate stability of principal magnetization and of the frugal sampling scheme (generally three hand samples per site), a frugality dictated by logistics and the need for wide coverage in this initial study. The fundamental dispersion question could be addressed in any renewed study of the volcanic sequences, this time taking, say, five or six hand samples per site to increase the accuracy of the site-mean directions. Such a detailed study, carefully planned, should be capable of addressing the paleosecular variation question and hence of contributing substantially to solving this long-standing tectonic problem. Such a renewed study would also allow a more accurate identification of the younger *M*-sequence anomalies in these sequences.

Acknowledgments

We wish to acknowledge the assistance of Alan Johnson of the University of Victoria and Karen Wallace-Dudley of the Institute of Sedimentary and Petroleum Geology, who helped make the collections. The fieldwork was carried out under the auspices of the Polar Continental Shelf Project, whose officers were a tremendous help to us with the logistics of working in this difficult terrain. The pilots of both fixed-wing aircraft and helicopters earned our complete trust with their quiet competence. The staff of the Atmosphere Environmental Service weather station at Eureka were generous with their hospitality. Reviews by Charlotte Jackson and R. F. Butler were most helpful. To all these people go our warmest thanks.

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